

Does Gerrymandering Distort Measures of Racial Vote Polarization?

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Abstract

Academic and legal debates over racially polarized voting (RPV) hinge on a quantity we fundamentally cannot observe: individual vote choice. To date, most approaches to estimating vote choice by racial group – including ecological regression (ER) and ecological inference (EI) – work down from aggregate vote totals and estimates of voting age population by group. While researchers and practitioners are often interested in measuring racial vote polarization within districts, little is known about how the manipulation of district boundaries affects measurement performance. Using both formal analyses and simulation experiments, we demonstrate that strategic redistricting may induce ecological correlations that *distort RPV* measurement in precisely those maps *most likely* to dilute minority votes. The key feature of our theory is that a district’s racial composition serves as a signal of the decisive voters’ partisan attachments. If a mapmaker is not aligned with the minority-preferred (opposition) party, she has an incentive to place her party’s strongest supporters in competitive high-minority-share districts, where outcomes are less certain, rather than in similar low-minority-share districts. Using simulation experiments with varying levels of residential segregation, we show how pairing areas with low opposition support and low minority share alongside areas with relatively higher opposition support and minority share can bias ecological correlations. Our simulations further demonstrate that this bias cannot be attributed to assumptions about voters’ preferences, the information available to mapmakers or political geography. We provide the first evidence that partisan redistricting impacts measurement strategies for RPV and may carry far-reaching, unexpected implications for vote-dilution claims.

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1 Introduction

Secret ballots have long been understood as a fundamentally important democratic tool because they limit opportunities for coercion and voter intimidation (Goodwin-Gill, 1994; Teorell, Ziblatt and Lehoucq, 2017). Accordingly, secret balloting has been adopted in elections around the world and across diverse contexts. While this protects citizen privacy and bolsters faith in electoral systems, it also limits researchers’ and practitioners’ ability to study group voting patterns. This makes it difficult to answer a host of questions such as whether preferences for candidates or ballot initiatives differ across racial groups or genders (Goodman, 1953; Greenland and Robins, 1994; King, 1997), or to assess the extent of ticket-splitting (Burden and Kimball, 2002).

To address these questions, researchers typically rely on one of three approaches. First, they may use surveys, though obtaining representative samples of small geographic areas is challenging and respondents may misreport their choices (Hanmer, Banks and White, 2013). Alternatively, individual voter files can provide information on demographics and party affiliation, but group identity (in particular, race) is often incomplete, registration data are only partially observed, and party registration does not guarantee a vote is cast for that party. The third and most prevalent approach uses ecological inference techniques to estimate the shares of voters in each group who might have either turned out or voted for a given candidate using aggregate candidate-level (or party-level) vote totals and eligible voters by group. These methods, which form the backbone of billions of dollars worth of litigation over racially motivated redistricting, include analyses of homogeneous areas, ecological regression (ER), and ecological inference (EI).

Each of these ecological techniques presents significant challenges because they rely on assumptions about spatial autocorrelation and aggregation bias that are frequently violated in the real world (Greenland and Robins, 1994; Achen and Shively, 1995; Tam Cho,

1998; Gelman et al., 2001). In the context of voting, practitioners must often assume that members of a certain group will exhibit the same turnout or vote choice regardless of the demographic composition of their precincts or districts. Yet scholars have provided robust evidence against this assumption. One example of this is “racial threat,” whereby the presence of one group of voters induces another group to alter their vote choices (Key, 1949; Enos, 2016; Jardina, 2019). Widespread racial residential segregation further exacerbates these violations. Prior research has typically acknowledged these violations and proceeded to produce ecological inferences for the relevant geographies - however segregated they may be.

Additionally, often out of necessity since voting districts, census blocks, and other geographic units that form the building blocks of a legislative map are fixed at the time researchers study them, prior research has implicitly taken these boundaries as fixed or exogenous without explicitly construing them as products of strategic choices by map-makers seeking to maximize their party’s seat share, protect incumbents, or any number of other goals. The existing literature has largely elided the concern that the strategic aggregation of precincts, blocks, and voter tabulation districts (VTDs) into legislative districts, too, constitutes a potentially serious violation of the assumptions required to accurately estimate group behavior from aggregate data. One reason for this lacuna may be the difficulty of characterizing the key counterfactual baseline (i.e., the expected degree of bias in the absence of strategic redistricting) when individual ballots are unobservable. Relatedly, researchers can rarely benchmark the impact of mapmaker intent against the effects of underlying political geography.

This paper relies on both formal methods and simulation experiments to address these challenges. Formal analyses allow us to isolate the conditions under which strategic aggregation is most likely to distort ecological measures, relative to a counterfactual

nonpartisan mapmaking process. In particular, we pay close attention to both features of the underlying political environment (such as partisan sorting, racial bloc voting, and racial segregation) and endogenous district characteristics (such as competitiveness and minority share). We show that ecological approaches will perform especially poorly in maps strategically manipulated for partisan gain, and the bias will be worse in highly competitive and racially segregated districts. This occurs because partisan mapmakers anticipate that allocating favorable voters to the most competitive seats generally yields a larger boost to the party’s win probability than adding those voters to safe districts (Gul and Pesendorfer, 2010; Sabouni and Shelton, 2022). Likewise, if minority voters reliably support the opposition party, allocating more friendly voters to high-minority-share districts (where the party’s electoral advantage is less certain) offers more value than doing so in low-minority-share districts. As the mapmaker allocates low-opposition-support, low-minority-share precincts to relatively high-opposition-support, high-minority-share districts, average support for opposition candidates may become markedly lower in low-minority-share precincts than in high-minority-share precincts.

Next, we simulate thousands of possible maps within a single, hypothetical state. Unlike real-world elections, a simulation-based approach allows us to observe “true” vote choice at the individual level; we can use this “ground truth” to test how well state-of-the-art ecological methods actually perform under varying levels of aggregation bias and residential segregation. In addition, our simulation experiments permit us to isolate the causal effects of strategic redistricting on measurement performance across a broader range of data-generating processes than would be possible in theory. Finally, they also allow us to benchmark the effects of mapmakers motivations against political geography.

Our primary results focus on how well ecological regression (ER) and ecological inference (EI) recover true within-group vote shares across simulated redistricting plans that

reflect different mapmaker incentives and constraints. Our baseline simulations are designed to resemble a state demographically and politically comparable to North Carolina but without racial residential segregation. Relative to that baseline, ecological estimates tend to deviate from the true spread (minority - majority Democratic support) by up to six-seven percentage points in simulated environments with greater racial residential segregation and aggregation bias. More importantly, we show that with high levels of racial segregation, existing approaches consistently *underestimate* racial vote polarization in those districts *most likely* to dilute minority votes: competitive districts in the most Republican-leaning maps. The difference in performance between the most and least gerrymandered plans is roughly one-tenth as large as the difference in performance between the lowest- and highest-segregation simulations.

In sum, our findings provide a theoretical and simulation-based framework for researchers and litigants to diagnose measurement challenges arising from the strategic manipulation of district boundaries. In doing so, these findings complement research recommending augmenting ecological techniques with individual-level survey data ([Glynn and Wakefield, 2010](#); [Decter-Frain et al., 2025](#); [Kuriwaki et al., 2024](#)). More broadly, our findings suggest that strategic aggregation may distort ecological estimates beyond the electoral context: whenever decision-makers organize aggregate units around an institutionally salient group, later inference will partly recover the assignment rule, not just group behavior.

2 Background

2.1 Measuring Racially Polarized Voting (RPV)

From the perspective of measurement, researchers in an ideal world would just record every individual voter's race and vote choice, aggregate the individual data up to vote shares by candidate (or party) and race, and report the difference in a given candidate's vote share among white and non-white voters as a measure of racially polarized voting (RPV). In a world with two groups and two political parties, R and D , and this might look something like:

$$\text{Vote } D_i = \beta_{jd}^w + \Delta_{jd} \cdot x_i + \epsilon_i$$

where $x_i = 1$ is an indicator for voters belonging to the racial minority group and $\text{Vote } D_i = 1$ is an indicator for Party D supporters among voters (indexed by $i=1,\dots,n_j$), residing in precinct j and district d . Here, β_{jd}^w represents the share of non-minority (majority) voters in each precinct who vote for Party D candidates. $\Delta_{jd} \equiv \beta_{jd}^w - \beta_{jd}^m$ is the average difference in support for Party D candidates between majority and minority voters. Since our interest is in aggregation bias (not selection bias), we assume that the degree of racial vote polarization varies across precincts due to unobservable factors (ϵ_i) that are uncorrelated with race.¹ Though there is no specific threshold prescribed in the law, sufficiently large values of district-level RPV imply that majorities of minority and non-minority voters prefer different candidates.

As the secret ballot prevents direct estimation of Equation 1, legal experts and researchers are left to estimate the shares of white and non-white voters who might have voted for a given candidate from aggregate totals of votes by candidate (or party) and el-

¹ The Gingles decision, discussed below, explicitly forecloses inquiries about causation: "...it is the difference between the choices made by blacks and whites not the reasons for those differences – that results in blacks having less opportunity than whites to elect their preferred representatives." (Thornburg v. Gingles, 478 U.S. 30 (1986)).

eligible voters by race. In 1986, the Supreme Court affirmed that two statistical techniques for estimating within-group vote shares from aggregate data were “standard in the literature for the analysis of racially polarized voting” (Thornburg v. Gingles, 478 U.S. 30 (1986), [Greiner, 2007](#)). The first of these techniques is called homogeneous precincts analysis ([Grofman, Handley and Niemi, 1992](#)). This typically involves calculating vote shares in precincts where one racial group constitutes 90% or more of the residents and assuming total vote shares in that jurisdiction (1) accurately proxy for the preferences of the dominant group and (2) can be substituted for that group’s preferences in other locations. The second technique - ecological regression (ER) - remains the most widely used approach to estimating racially polarized voting from aggregate data ([de Benedictis-Kessner, 2015](#); [Greiner, 2007](#)). Researchers using ER estimate an ordinary least squares model in which they regress vote share for a particular candidate or party on each jurisdiction’s racial composition:

$$D \text{ Voteshare}_{jd} = \alpha_{eco}^0 + \Delta_{eco} \bar{x}_{jd} + \epsilon_{jd} \quad (1)$$

The key identifying assumption in Equation 1 is that voters within each group are similar enough that β_{jd}^b and β_{jd}^w can be assumed to be constant across all precincts ([Goodman, 1953](#)). Ecological Inference (EI), introduced by ([King, 1997](#)) is the second most common tool used to estimate within-group vote shares and diagnose the presence of RPV in litigation over districting plans ([de Benedictis-Kessner, 2015](#)). Among other differences, EI replaces the constancy assumption in Equation 1 with the assumption that β_{jd}^b and β_{jd}^w follow a known parametric distribution e.g., the bivariate truncated normal distribution.²

²[s]ince most areal units were created for administrative convenience and aggregate individuals with roughly similar patterns, ecological observations from the same data sets usually do have a lot in common. Even though Goodmans assumption that [the parameters] are constant over i is usually false, the assumption that they vary but have a single mode usually fits aggregate data (King 1997, 191192).

Ecological methods for estimating district-level RPV have been applied at three different geographic scales. Some analysts seek to draw inferences about specific districts from models fit to precincts across all districts in a state or jurisdiction ([King, 1997](#)). While convenient, this across-district approach can obscure district-level heterogeneity that is legally and substantively consequential. A second approach estimates polarization within intermediate administrative units, such as counties ([de Benedictis-Kessner, 2015](#)). Although this strategy avoids reliance on assumptions of statewide homogeneity, it has been criticized for hinging on arbitrarily defined boundaries and for permitting cherry-picking of favorable geographic units. Recognizing these limitations, a third approach—now common in litigation and expert practice—conducts RPV analysis district by district, using only the precincts within each district (see e.g. [Stephanopoulos, McGhee and Warshaw \(2023\)](#)). In their review of district court opinions, [Elmendorf and Spencer \(2015\)](#) found that forty cases relied on polity-wide analyses, twelve on district-level analyses, and five on regional (e.g., countywide) analyses. Our paper focuses on this third approach.

2.2 Implications of Redistricting on Aggregation Bias

Our main point of departure from existing research on estimating district-by-group-level outcomes from aggregate data is the explicit inclusion of mapmakers with partisan intentions. To evaluate the consequences of strategic redistricting on model violations, consider the standard decomposition of aggregation bias (c.f. [Greenland and Morgen-](#)

stern (1989); Glynn et al. (2008)):

$$E[\bar{\Delta}_{eco,d}|\bar{x}_d] - \bar{\Delta}_{n,d} = \underbrace{\frac{Cov_n(x, \beta^w)}{Var_n(x)}}_{\text{Intercept Bias}} + \underbrace{\bar{x}_d \frac{Cov_n(x, \Delta)}{Var_n(x)} + \frac{Cov_n((\bar{x} - \bar{x}_d)^2, \Delta)}{Var_n(x)}}_{\text{Slope Bias}} \quad (2)$$

where $E[\bar{\Delta}_{eco,d}|\bar{x}_d]$ is the expectation of the ER estimate conditional on precinct minority shares ($\bar{x}_d$) and $\bar{\Delta}_{n,d} \equiv \sum_{j \in S_d} \Delta_j$ is the average (population-weighted) difference in vote shares by race within each district.³ The first term on the right hand side of Equation 2, which Glynn et al. (2008) refer to as *Intercept Bias*, captures the weighted least-squares slope from the regression of non-minority voters' political behavior (β^w) on precinct minority share. This term would be positive if, for example, non-minority voters living in or near heavily minority communities share their neighbors' economic, social, or political preferences and therefore vote similarly. Conversely, the term would be negative if white voters living near concentrated minority communities respond with racial threat, becoming more likely to vote cohesively against the interests of their non-white neighbors because proximity heightens the perceived need to protect their own group interests (Jardina, 2019; Enos, 2016; Key, 1949). The second and third components of Equation 2 (collectively referred to as *Slope Bias*) represent the bias induced by the covariance between racial composition and racial vote polarization across precincts in a district.

³Our empirical analysis of district RPV focuses on a closely related parameter: the (group-weighted) difference between the minority support rate and the majority support rate

$$\bar{\Delta}_d \equiv \underbrace{\frac{\sum_{j \in S_d} n_j \bar{x}_j \beta_j^b}{n_d \bar{x}_d}}_{\text{Minority Party D Rate}} - \underbrace{\frac{\sum_{j \in S_d} n_j (1 - \bar{x}_j) \beta_j^w}{n_d (1 - \bar{x}_d)}}_{\text{Majority Party D Rate}},$$

where $\bar{x}_j \equiv \frac{1}{n_j} \sum_{i=1}^{n_j} x_{ij}$ is the share of minority voters in precinct j , and $n_d \equiv \sum_{j \in S_d} n_j$ is the total district population. We prefer this estimand because it clarifies the role of residential segregation on measurement performance. Residential segregation shapes mapmakers' choices and can, thus, influence substantive outcomes for minority groups. See Appendix B.1 for details.

Equation 2 implies that the within-district ER estimator ($\hat{\Delta}_{eco,d}$) is especially vulnerable to strategic redistricting because small changes in the composition of precincts can dramatically alter the covariance between racial composition and group voting behavior. In the following section, we develop a simple heuristic model that illustrates the political and geographic conditions under which strategic redistricting is most likely to affect precinct composition relative to a non-partisan map-making process.

2.3 Relationship to Literature

A series of previous investigations have approached the challenge of estimating group-level outcomes from aggregate data with different goals: diagnosing and bounding the potential damage done to accuracy if assumptions are violated (King, 1997; Tam Cho, 1998); comparison and validation to voter files, exit polls, or other survey data (Grofman and Barreto, 2009; de Benedictis-Kessner, 2015); augmenting estimation techniques (Glynn and Wakefield, 2010; Decter-Frain et al., 2025; Kuriwaki et al., 2024), and diagnosing issues that arise when researchers attempt to expand these methods beyond the two-group, two-party case (Ferree, 2004; Barreto et al., 2019). We contribute to this literature by explicitly characterizing measurement outcomes under extremely partisan and neutral maps. We base our analysis on simulations of entirely synthetic electorates where individual vote choice is observed. We describe the parameters of our simulations in detail below, but this allows readers to contemplate a specific state with a fixed population and constraints on possible districting. We focus on the two-group, two-party case. Our simulations incorporate varying levels of racial residential segregation and two sources of model violations – spatially correlated random coefficients (Anselin and Cho, 2002; Achen and Shively, 1995) (to capture the effects of unmeasured similarity of voters in an area) and a correlation between model parameters and minority share (to

capture the effects of aggregation bias).

3 Partisan Mapmakers and Minority Voters

In this section, we present a heuristic model of legislative redistricting in the presence of racially polarized voting and extreme residential segregation. Our model is closely related to “crack-and-pack” theories of partisan gerrymandering which predict that motivated mapmakers differentiate between districts likely to favor their own party and those likely to favor the opposition, carving out narrow but safe aligned majorities in the former and overwhelming majorities of unaligned voters in the latter (Owen and Grofman, 1988; Gul and Pesendorfer, 2010; Sabouni and Shelton, 2022)⁴.

Given a status quo plan, a mapmaker can compare the marginal benefit of swapping any pair of precincts between adjacent districts. Whereas a neutral (non-partisan) mapmaker will consider traditional redistricting criteria – such as compactness, constitutional protections for communities of interest or respect for traditional political subdivisions – a partisan mapmaker may also consider policy and office-holding motivations. To fix ideas, we begin by reviewing strategic redistricting in an environment where segregation and racial vote polarization are absent. Following Gul and Pesendorfer (2010), a partisan mapmaker would prefer to allocate aligned voters to competitive party seats rather than safer ones because the party’s probability of winning increases in a district’s partisan composition but at a decreasing rate.

Next, we discuss the optimal allocation of voters across districts in an electoral environment of racial vote polarization. Specifically, we evaluate how the anticipated electoral benefit of strategic redistricting varies across districts with similar partisan compositions but different racial compositions. Our key finding is that, under natural assump-

⁴Cox and Holden (2011), but also see Kolitilin and Wolitzky (2024)

tions, the anticipated electoral value of allocating favorable voters to a district increases with the share of minority voters in that district. Consequently, strategic redistricting induces a greater correlation between minority population share and within-group preference heterogeneity than would be expected from a nonpartisan mapmaking process. Finally, we show that the aggregation bias induced by strategic redistricting leads ecological methods to *overestimate or underestimate* racial vote polarization in precisely those districts *most likely* to dilute minority votes.

3.1 Preliminaries

a. Individual vote choice. There is a continuum of voters spread across N equally-populous single-member districts, indexed by $j \in \{1, \dots, N\}$. Voters belong to one of two racial groups, labeled $g \in \{w, m\}$. Let β_j denote the fraction of voters within district j who belong to the racial minority group, m . The remaining $1 - \beta_j$ voters in district j belong to the majority group. Each voter must cast their ballot for either one of two parties, D and R . In doing so, voters considers three sets of factors: (i) their policy preferences, (ii) their degree of identification with party D if they belong to the racial minority group, and (iii) aggregate electoral shocks favoring one party over the other. We describe each of these factors below.

First, we assume voters have symmetric, single-peaked preferences over a unidimensional policy space with support from $[-1, 1]$. Voter utility depends, in part, on the distance between the voter's ideal point (denoted by θ) and the implemented policy. We take party positions as given: party R policy is fixed at 1, and party D policy is fixed at -1. Voters whose ideal points are above zero are referred to as "affiliated with" or "belonging to" party R , while voters with ideal points below zero are affiliated with party D . Let π denote the share of voters affiliated with party R . We assume these voters'

ideal points are drawn from a cumulative distribution I_R , which is continuous, strictly increasing, and convex on the unit interval, with $I_R(0) = 0$ and $I_R(1) = 1$. Likewise, the remaining $1 - \pi$ voters, affiliated with party D , have ideal points drawn from a cumulative distribution with symmetric properties: $I_D(\theta) = 1 - I_R(-\theta)$ for all $\theta \in [-1, 0]$.

To capture racial vote polarization, we assume party D is a minority-preferred party in the sense that some minority voters gain utility from supporting candidates of that party for reasons beyond immediate policy.⁵ This assumption aligns with empirical evidence that political norms in minority communities foster partisan unity despite diverse individual preferences (White and Laird, 2020). For example, minority voters strongly identified with their group may find that coordinating their support around one party is the best means to protect their voting rights or influence national policy (Tate, 1994; Dawson, 1995). This assumption is also consistent with a preference for co-ethnic candidates (or discrimination against non-co-ethnic candidates) who overwhelmingly sort into one party over the other (Fraga, 2016). While competing accounts of racially polarized voting imply different substantive interpretations – and possibly different legal remedies – our focus is on the empirical implications of a divergence in voting behavior across groups, rather than the origins of that divergence.

Specifically, we assume that minority voters enjoy an idiosyncratic benefit, δ_i , whenever a party D candidate is elected in their district. δ_i is uniformly distributed on the interval $[0, 2\delta]$, where larger values of $\delta \geq 0$ indicate greater racial bloc voting.⁶

Finally, an aggregate valence shock, $\nu \in \mathbb{R}$, may further tilt the district’s electorate toward one party over the other. Positive realizations of the valence shock favor the party D candidate. We assume that ν is drawn from a cumulative distribution, L , which

⁵The assumption of bloc voting among only minority voters – rather than assume racial bloc voting within either groups – is without loss of generality.

⁶Note that this assumption allows for, but does not assume, group-based differences in average support for the typical policies or social choices associated with either party.

is strictly concave on $\mathbb{R}_+$, continuous and symmetric around zero.

b. Legislative redistricting: Article I, Section 4 of the U.S. Constitution grants state legislatures the responsibility of drawing redistricting maps. State legislatures retain primary control over congressional redistricting in thirty-nine U.S. states and state legislative redistricting in thirty-five states.⁷ For simplicity, we consider a unitary partisan mapmaker (the majority party) with control over a single state.⁸ The party must determine the change in party R vote-share (which we denote Δ_π) within each district, subject to the constraint that the statewide partisan alignment is held constant (i.e., the net shift in R vote-share across districts, $\sum_{j=1}^N \Delta_{\pi j}$, is zero). Throughout, we assume that the mapmaker is constitutionally forbidden from targeting minority or non-minority voters to reassignment to other districts.

While the party lacks information about individual voter preferences, it can observe the typical party R share of the normal vote (or share of registered voters) in each voting precinct. The mapmaker may, for example, average the most recent Presidential election returns within each precinct. We assume the party has both policy and office-holding motivations. Thus, as the party decides how to allocate a pool of favorable or aligned voters across equally-sized districts, they must balance the majority party's policy motivations (as proxied by the party's overall seat share) with each member's re-election concerns.

⁷See [All About Redistricting](#).

⁸Specifying a fully micro-founded legislative bargaining game between members of the majority party is challenging as the intra-party negotiations leading to a legislative map are typically unobserved. For an axiomatic solution to legislative bargaining over redistricting, please see [Sabouni and Shelton \(2022\)](#). Their characterization of the equilibrium distribution of voters across districts (without regard to race) is consistent with logic of strategic redistricting described below.

3.2 Strategic Redistricting in Racially Polarized Electorates

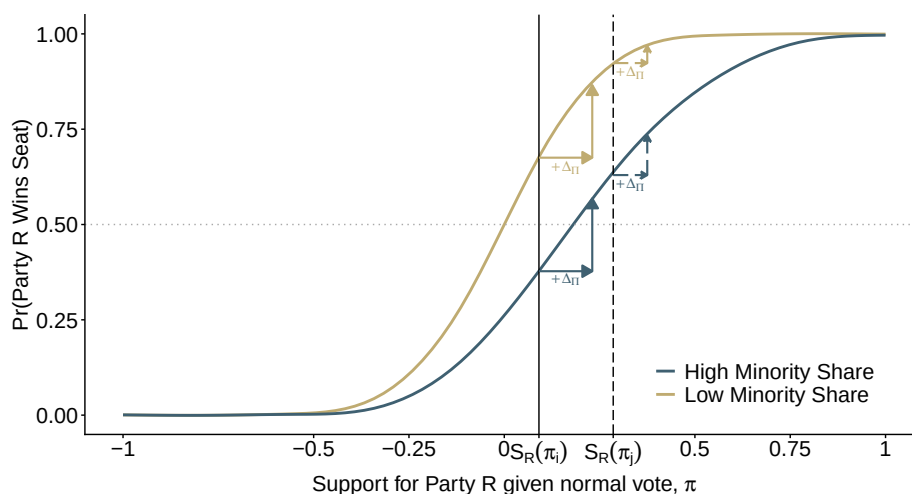
In this subsection, we demonstrate that a party R -biased mapmaker will be more likely than a nonpartisan mapmaker to allocate high party R share precincts into districts with *relatively high minority share* than into comparable districts with *relatively low minority share*. More specifically, we argue strategic redistricting will induce a correlation between the percentage of minority voters and party D vote share in (competitive) districts where the party's electoral advantage is less certain. Second, we illustrate conditions under this correlation will induce more or less bias in ecological estimates of RPV than would be expected given the spatial distribution of voters.

The intuition behind the first step of our argument is closely related to a standard argument about strategic redistricting under electoral uncertainty: mapmakers will, *ceteris paribus*, allocate more aligned voters to competitive seats than uncompetitive seats. Figure 1 illustrates this point using a series of simulated districts. The x-axis depicts the location of the pivotal voter's ideal point in each district, and the y-axis corresponds to party R 's likelihood of electoral success. We assume electoral shocks can dramatically affects election outcomes, but only in districts where vote support for both parties is balanced. The far left corner, for example, represents a district composed entirely of party D voters and where party R candidates are virtually unelectable.

To fix ideas, consider two hypothetical districts in the legislative majority with the same racial composition but different party R vote shares ($\pi_j > \pi_i > 0.5$) prior to redistricting. The pivotal voter's ideal point in district j (denoted by a dashed vertical line) falls to the right of the median voter's ideal point in district i (denoted by solid vertical line). The solid black curve represents party R 's probability of winning if minority share (or racial vote polarization) is minimal. This probability is increasing in the pivotal voter's preference for party R policy, but at a decreasing rate. As a result, the marginal

increase in party R 's probability of winning after increasing its vote share by $\Delta\pi$ percentage points (vertical arrows) is, all else equal, greater in competitive districts than in safe ones.

Figure 1: Strategic Redistricting in Racially Polarized Electorates



See text for details. The valence shock is drawn from $\mathcal{N}(0, 0.2)$. The extent of minority identification with the D party is drawn from $U(0, 0.6)$.

Next, consider two party- R districts with the same partisan makeup prior to redistricting but different racial compositions: one is 65% minority, the other 15%. If minority bloc voting is minimal, both districts will receive an equal number of R voters, as shifting votes yields roughly the same shift in party R 's probability of winning in either district. If minority bloc voting is significant, party R can expect to fare worse in the high-minority share district (solid gray curve in Figure 1), where the minority group's attachment to party D may be decisive, than in the less diverse district (solid black curve). Hence, the marginal impact of adding aligned voters on party R 's probability of winning will be greater in the former seat than in the latter. Moreover, as Figure 1 indicates, this marginal impact is increasing in district competitiveness: whereas allocating $\Delta\pi$ friendly votes to

a competitive high-minority-share district (e.g. with vote share π_i) may flip the seat in party R 's favor, the same allocation in an equally diverse but less competitive seat (e.g. with vote share $\pi_j > \pi_i$) would not.

We present formal results characterizing the partisan mapmaker's optimal redistricting decisions in Appendix A. In particular, we show that, in an environment without residential segregation, a net increase in party R voters shifts the location of the median voter within the minority group rightward, reducing the influence of the minority group's attachment to party D on district election outcomes. As a result, a mapmaker seeking to maximize R -friendly policy will prefer to assign reliable party R precincts in high-minority-share districts – where the party's electoral advantage is less certain – over comparable low-minority-share districts.⁹

Finally, we examine strategic redistricting in an electoral environment with racial vote polarization and aggregation bias (i.e., correlations between minority share and local partisan preferences). Consider, for example, a racially polarized electoral environment in which like-minded and co-ethnic voters cluster together in space. Geographic sorting by party affiliation has risen sharply in the past fifty years and appears across all geographic levels (Brown and Enos, 2021; Kaplan, Spenkuch and Sullivan, 2022), and is an fundamental source of within-group preference heterogeneity that may induce aggregation bias. Allocating Δ_π aligned voters to high minority share districts can serve two distinct purposes for party R mapmakers facing this type of electoral environment. First, it can dilute the impact of minority voters' partisan attachments on electoral outcomes. Second, it can lower the minority's population share, reducing the probability that minority voters are pivotal. As these effects reinforce one another, partisan sorting and residential segregation exacerbate the mapmaker's incentive to prioritize allocating friendly voters

⁹For simplicity, we implicitly assume that re-assigning the precinct would not affect other traditional redistricting criteria, so that a nonpartisan mapmaker would have no reason to re-assign the precinct.

to districts with the high minority share seats before those with low minority share.

3.3 Implications for Measurement Performance

To illustrate the implications of strategic redistricting on measurement performance, we simulate a hypothetical district with 120 precincts using the same parameters as in Figure 1. We assume that the district is relatively competitive and has a sizable minority population that is unevenly distributed across precincts.¹⁰ The top left panel of Figure 2 plots the true share of party *D* voters among minority voters (blue triangles) and majority voters (yellow triangles) against minority share in each precinct. Vote choice is polarized by racial group, but group voteshares are not correlated with minority share. The bottom left panel depicts the total party *D* voteshares across simulated precincts. The upward sloping dashed line is an OLS ecological regression (ER) line. Ecological Regression (ER) closely approximates the true spread in this example (approximately 46-49% percentage points), while Ecological Inference (EI) underestimates RPV by over 25% percentage points.

Now, suppose the mapmaker has an incentive to prioritize allocating party *R* voters to districts with the high minority share seats before those with low minority share. As low party *D* support precincts are paired alongside precincts with relatively *higher average party D support and higher average minority share*, the correlation between the percentage of minority voters and party *D* vote share (represented by Δ_{eco} in Equation 1) rises. The top right panel of Figure 2 plots the true share of party *D* voters among minority voters (blue triangles) and majority voters (yellow triangles) within the original district and the reassigned precinct. Dashed lines represent the linear relationship between vote

¹⁰Voters' ideal points are sampled from a normal distribution with mean = 0 and standard deviation = 0.025m while precinct minority share is sampled from a truncated normal distribution with mean = 0.4 and standard deviation 0.05.

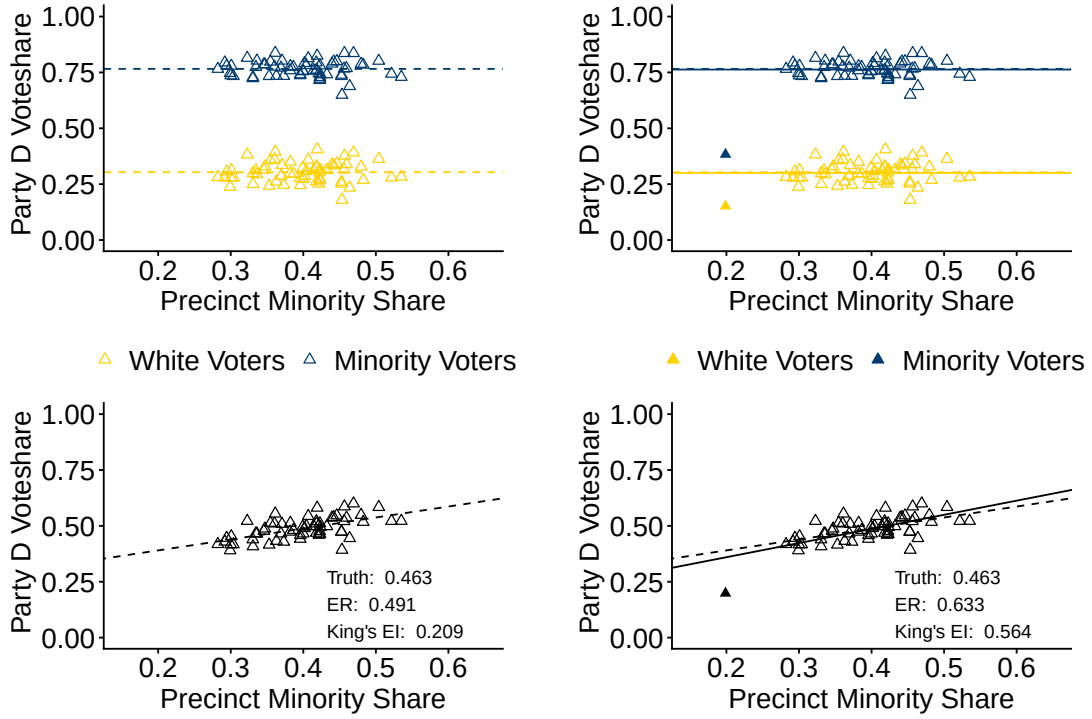


Figure 2: Illustration of how race-conscious redistricting can affect measurement strategies for racially polarized voting (RPV), including Ecological Regression ("ER") and Ecological Inference ("King's EI"); reassigning a precinct (solid dot) with relatively low party-*D* share and low minority share distorts observed ecological correlations within the district without affecting RPV.

share by group and minority share within the original district, while solid lines indicate corresponding values after the precinct's reassignment.¹¹ As the bottom right panel of Figure 2 shows adding a precinct with unusually low party *D* vote share and minority share (red dot) can increase ER/EI estimates by 14-40% percentage points, even though true racial vote polarization is unchanged.

Under what conditions will mapmakers' incentives confound ecological estimates of RPV? In Appendix A.3.3, we generalize this example to evaluate the impact of reassign-

¹¹Notice that relative to the original district, majority-group support for party *D* and the gap between majority and minority support are halved in the new precinct.

ing a hypothetical precinct to a district with higher minority share and party D vote share. We show that the precinct reassignment's can either increase or reduce ecological bias, depending upon the degree to which those ecological biases appear across precincts within the district, and between the district and the marginal precinct. This last finding implies that the implications of mapmakers' incentives on measurement performance will vary depending upon the degree of aggregation bias within the state.

4 Data and Empirical Approach

To more systematically investigate the impact of redistricting on measurement, we conduct a series of simulation experiments that vary the sources and magnitude of aggregation bias. Our baseline simulations are designed to capture the key political, demographic, and spatial characteristics of North Carolina, including overall partisanship, racial composition, urbanization, and residential segregation. We then systematically modify certain features to generate a range of electoral environments in which standard ecological inference is likely to fail. Finally, we use state-of-the-art redistricting algorithms to construct maps that satisfy traditional redistricting criteria as well as maps that maximize explicit partisan objectives. Following the argument in the previous section, our partisan maps are designed to aggressively maximize expected seat share, recognizing that districts with similar compositions can differ in their predictability based on district racial composition and the degree of racial vote polarization. This approach allows us to isolate the extent to which ecological bias can be attributed to mapmaker choices rather than the underlying political and demographic environment in which districts are drawn.

4.1 Simulation Experiments

Our simulation procedure comprises four main steps, which are summarized here. A more detailed description of the simulation workflow is provided in Appendix C. The process begins with a set of atomic units (indexed by $j = 1, \dots, J$), each characterized by a total population (N_j) and a minority population (X_j). These units may represent voting districts (VTDs) from the Decennial Census’s PL 94-171 file, precinct shapefiles with spatially interpolated population counts, or sets of population and minority population values drawn from a specified distribution. For the results presented below, we use a fully simulated population of approximately two million voters distributed across 830 electoral precincts and two urban centers, with demographic characteristics (population counts and minority population shares) loosely based on the 2020 North Carolina VTDs from the PL 94-171 file.¹² At this stage of the simulation, the population consists of individual voters, each assigned a race and allocated to a precinct.

Next we expand the population counts and assign individual votes (Democratic or Republican) using EI’s multivariate truncated normal model with parameters for the average and variance of each group’s voting behavior, as well as the correlation between the groups’ voting behavior. These units are atomic in the sense that from this point forward, each individual’s minority status, observed vote, and membership in their unit remains fixed. Variation enters in how these groups are placed on a map and how districts are drawn to combine different units together.

We then create the spatial geography of our simulated state, by re-assigning these units to a grid varying the assignment mechanism from one that is completely uncorrelated with minority status and vote behavior to one that is strongly determined by a weighted combination of the two. We consider two distinct levels of segregation. In the

¹²A full list of simulation parameters is provided in the appendix.

low segregation scenario, Moran’s I – a measure of spatial autocorrelation – is consistently near zero for the precinct share of minority voters and democratic voters; in the high segregation scenario, Moran’s I is approximately 0.8 and statistically distinguishable from zero.

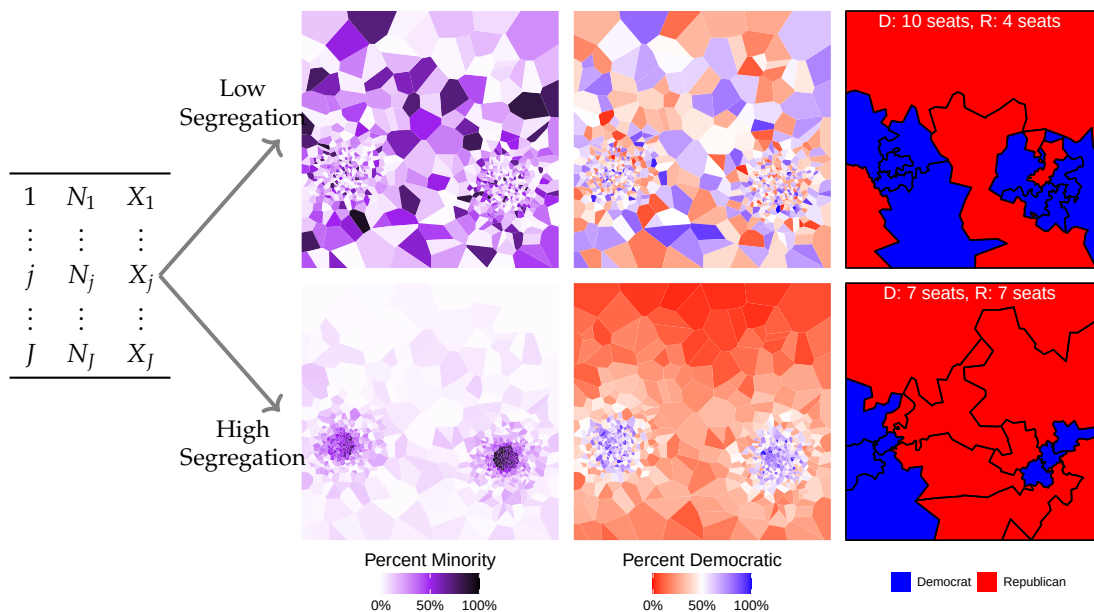


Figure 3: Simulation Procedure. We generate a population of 1 million voters (with 25% minority share) distributed across 830 electoral precincts. We implement King’s method for Ecological Inference, by jointly drawing correlated minority and majority voting probabilities within precincts from a truncated normal distribution. Next, we assign that fixed population to a spatial grid with either low or high residential segregation. In the final step, we generate maps that sorts those voters into 14 roughly equal population districts.

Finally, we use computational redistricting methods (DeFord, Duchin and Solomon, 2021) to draw ensembles of legislative maps that a mapmaker might produce under different objectives: a neutral objective, a partisan Democratic objective, and a partisan Republican objective. We simulated extreme Republican and Democratic maps by optimizing each party’s expected seat share, taking into account racial composition and

racially polarized voting patterns. To capture race-conscious partisan gerrymandering, we incorporate the effects of racial composition and polarization on both expected partisan support and electoral uncertainty, recognizing that districts with similar partisan composition can differ markedly in their predictability. Specifically, the expected seat share is calculated by summing up the party’s probability of winning each district, which we assume takes the following form $Prob(Win_d) = \Phi(\frac{\mu_d - 0.5}{\sigma_d})$, where μ_d and σ_d are the average and standard deviation of the normal voteshare in district d .¹³

Our simulation approach yields a 2×3 typology of redistricting ensembles, each categorized by electoral environment (low or high segregation) and a mapmaker objective (Democratic, neutral, Republican) with 1000 maps in each cell. We fit different ER and 2x2 EI models for each of the 14 simulated districts in each simulated electorate, using only the precincts within each district and storing point group-specific estimates of district-level parameters.

4.2 Main Results

Panels A and B of Table 1 summarize performance comparisons for ER and EI respectively. We report the root mean square error (RMSE) for both measures along with the bias, or average magnitude of the differences between the true and predicted majority and minority Democratic shares by district. In the context of litigation, the relevant question is often whether or not a particular method detected the presence (or absence) of RPV correctly. We report this as well, defining RPV as just an indicator for whether the majority and minority group vote shares for the Democratic candidate fall on opposite sides of 50%, as in Kuriwaki et al. (2024) and many expert reports on this topic. Readers interested in how district- and map-level electoral characteristics vary across redistricting

¹³Relative to an optimizer that simply maximizes vote shares, maximizing expected seat share under electoral uncertainty encourages the dispersion of reliable out-party voters across multiple districts.

and segregation scenarios can refer to Appendix Table 5.

To benchmark the impacts of strategic redistricting on measurement performance, we begin our discussion of Table 1 by comparing the performance of standard ecological estimates across simulated neutral (i.e., non-partisan) maps for each of the electoral environments. Both ER and EI recover minority and majority Democratic support in simulations with little correlation between group sizes and vote support by group (row 2 of Panels A and B), correctly classifying RPV in 96-99% of districts. ER tends to misreport the true difference between minority and majority Democratic support (-2.3 percentage points) compared to EI (-1.1 percentage points). Relative to the low-bias scenario, we observe a *six- sevenfold* increase in the root mean square error for minority Democratic share in the worst-bias scenario (fifth row in each panel of Table 1). Both ER and EI consistently over-estimate majority Democratic support by 6-8 percentage points while under-estimating minority Democratic support by as much as 18-29 percentage points. In sum, EI underestimates by 25 percentage points while ER underestimates RPV by 38 percentage points in our simulated environment. As a consequence, both approaches correctly identify districts with RPV only 55-67.5% of the time and actually flip the sign on the difference between the shares of true majority and minority voters who support the Democratic candidate between one-third to one-half of the time in the worst case scenario. Given systematic underestimation of minority vote shares in particular, this implies that these measures are particularly biased in detecting voting patterns among the group of key interest to plaintiffs and work considerably worse than guessing in the presence of segregation.

To what extent does redistricting impact performance, and how does that effect compare to the effects of spatial geography? To answer this question, we report performance comparisons for Democratic and Republican maps alongside the non-partisan maps for

Table 1: Performance Comparison of Ecological Regression and Ecological Inference (Simulated States with Low or High Segregation)

Bias	Plans	RMSE		Bias			Ground Truth		Classification		
		Min.	Maj.	Min.	Maj.	Spread	Spread	True RPV	RPV Correct	Sign Flips	Pct. Under
Panel A: Ecological Regression (ER)											
Low	Dem.	6.9	2.3	-1.7	0.5	-2.3	38.6	100.0	99.1	0.0	62.0
	Net.	6.5	2.2	-1.8	0.5	-2.3	38.6	99.6	96.4	0.0	61.2
	Rep.	6.7	2.3	-1.9	0.6	-2.5	38.5	94.0	95.2	0.0	62.0
High	Dem.	39.4	11.5	-29.8	8.2	-37.9	36.9	80.0	58.6	47.0	93.2
	Net.	39.0	12.2	-29.4	8.7	-38.1	36.8	73.2	55.0	46.9	93.3
	Rep.	41.6	13.6	-33.5	10.2	-43.7	36.5	69.0	60.8	58.5	94.9
Panel B: Ecological Inference (EI)											
Low	Dem.	4.4	1.5	-1.0	0.3	-1.2	38.6	100.0	99.9	0.0	59.5
	Net.	4.1	1.4	-0.9	0.3	-1.1	38.6	99.6	99.5	0.0	60.9
	Rep.	4.5	1.5	-1.1	0.4	-1.4	38.5	94.0	97.5	0.0	59.8
High	Dem.	28.1	9.3	-17.7	5.8	-23.9	36.9	80.0	71.8	30.7	83.2
	Net.	28.0	9.9	-18.3	6.4	-25.1	36.8	73.2	67.5	32.2	85.8
	Rep.	29.7	11.1	-20.2	7.6	-28.4	36.5	69.0	75.5	39.3	84.9

Note: Performance metrics for ecological regression (ER) and ecological inference (EI) across different levels of segregation (Low and High) and redistricting scenarios (Democratic-favoring, Neutral, Republican-favoring). RMSE = Root Mean Square Error; Min. and Maj. refer to minority and majority groups, respectively. Under the Bias columns, Min. and Maj. show bias in estimating group-specific Democratic vote shares, while Spread shows bias in estimating the difference between minority and majority Democratic support. Ground Truth Spread is the average true difference Minority Dem % - Majority Dem %; True RPV shows the actual proportion of racially polarized voting in the data. RPV Correct indicates the proportion of correctly identified RPV cases. Sign Flips shows the proportion of cases where the estimated difference between majority and minority Democratic vote share has the opposite sign from the true difference. % Under shows the proportion of cases where ER or EI under-predict district-level minority voteshares for the Democratic candidate.

each simulated environment in Table 1. We find few differences between partisan and neutral maps in the low-bias environment: ER/EI estimates of the minority-majority Democratic support gap are only 0.2-0.3 percentage points smaller in Republican plans than neutral ones. While substantively small, this difference amounts to 8.6-27% of the typical bias in this environment.

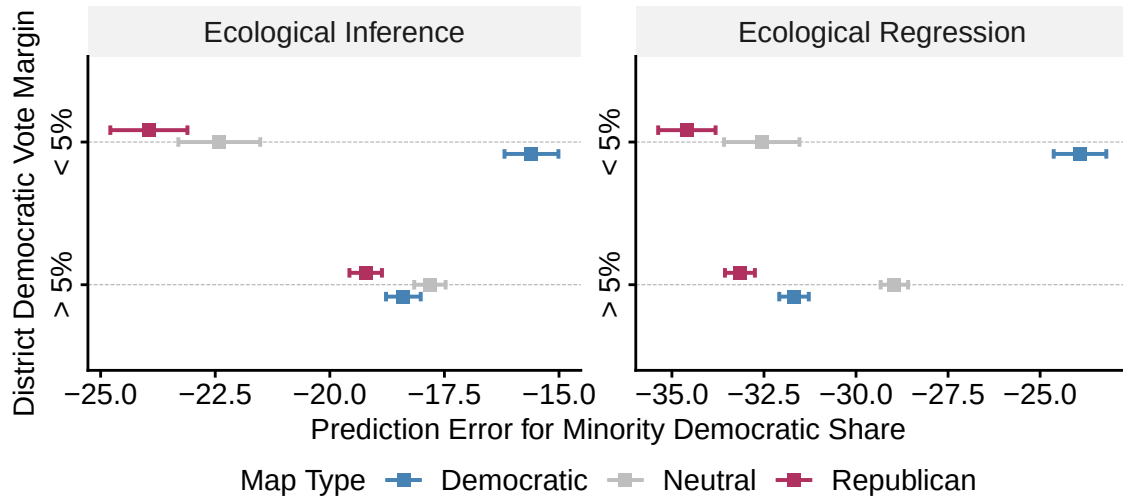
There are significantly larger differences between partisan and neutral maps when aggregation bias is high, with both ER and EI performing markedly worse in Republican maps. On average, the spread is underestimated by five percentage points more in Republican than in neutral maps. This difference is statistically significant and comparable in magnitude to 13-21% of the difference in measurement bias between the least- and most segregated electoral environments.

Turning to substantive outcomes, estimates from Republican maps are 10–12 points more likely to misclassify which of the two groups most favor Democratic candidates. Since ER/EI systematically underestimate racial vote polarization, both approaches tend to misclassify truly polarized districts as unpolarized. This, when combined with the fact that neutral plans have more racially polarized districts than Republican plans (73% vs. 69%) results in slightly lower overall accuracy for neutral maps (54%) than for Republican ones (59%).

4.3 Additional Results

Heterogeneous Effects One prediction of our theory is that strategic redistricting will distort ecological estimates most in competitive districts. To test this prediction, we fit simple OLS models predicting prediction error for minority Democratic voteshare. Our regression models include dummy variables for competitive districts (define as those where the district vote margin is less than five percentage points), mapmaker type and

Figure 4: ER/EI Prediction Errors by District Vote Margin (High Segregation)



Note: Predicted values from linear regression models for EI prediction errors (left panel) and ER prediction errors (right panel) of minority Democratic support. Underlying model estimates, and additional details, reported in Appendix Table 6.

their interactions (see Appendix Table 6). Figure 4 presents predicted values. We find relatively few partisan differences in EI (left panel) or ER (right panel) performance in uncompetitive districts, where minority Democratic support varies by just 0.5-4 percentage points depending on the mapmaker's type. We detect significantly larger differences in competitive districts, where minority Democratic support varies by as much as 8-11 percentage points.

Results using North Carolina's actual geography A key advantage of our simulation-based approach is that it can be modified to capture a range of assumptions about voter preferences or political geography. To highlight that advantage, we replicated our findings using the actual demographic compositions and spatial geography of the 2020 North Carolina VTDs. We directly applied the EI voter model in the previous section to generate statewide preferences for two groups (white and African American voters). This ex-

ample represents a realistic, best-case scenario for ecological inference, because, like the earlier low-segregation simulations, group preferences are largely concentrated around common statewide means. Appendix Table 7 presents key electoral features for this example across neutral, Democratic and Republican redistricting plans.

Table 2 presents performance metrics for both Ecological Regression (panel A) and 2x2 Ecological Inference (panel B) fit within each simulated district. Here, within-district EI tends to recover group-level parameters across mapmaker types (underestimating the spread by 4.4-4.8 percentage points), while ER model performs differently based on the mapmaker’s partisan objectives. In particular, ER tends to underestimate the spread in Republican maps (by 0.7 percentage points), while overestimating the spread in Democratic plans (by 0.4 percentage points), relative to neutral maps. While the differences across maps may appear small in absolute terms, they correspond to 8.33-14.5 percent of the typical measurement error in this best-case environment.

Table 2: Performance Comparison of Ecological Regression and Ecological Inference (North Carolina)

Plans	RMSE		Bias			Ground Truth		Classification		
	Min.	Maj.	Min.	Maj.	Spread	Spread	True RPV	RPV Correct	Sign Flips	Pct. Under
Panel A: Ecological Regression (ER)										
Dem.	11.7	2.0	-3.7	0.7	-4.4	31.4	92.8	84.2	6.7	65.7
Net.	12.0	2.1	-4.1	0.8	-4.8	31.3	91.7	83.4	6.9	68.0
Rep.	11.6	2.1	-4.4	1.0	-5.5	31.1	86.4	84.4	6.6	73.7
Panel B: Ecological Inference (EI)										
Dem.	5.1	1.2	-3.5	0.8	-4.4	31.4	92.8	93.7	0.0	85.1
Net.	5.3	1.2	-3.8	0.9	-4.7	31.3	91.7	93.7	0.0	88.9
Rep.	5.2	1.3	-3.8	1.0	-4.8	31.1	86.4	93.1	0.0	91.0

Note: Performance metrics for ecological regression (ER) and ecological inference (EI) using North Carolina’s actual geography. See Table 1 for more details.

5 Practical Recommendations

This section provides practical recommendations for ecological inference in gerrymandered districts and suggests directions for future research. The primary source of bias in this paper is a form of modifiable-area-unit instability: selecting aggregation boundaries to achieve partisan goals introduces noise into the resulting inferences. The standard recommendation in the literature for this type of problem is statewide ecological inference (King, 1997), estimating group parameters using a single model across the entire state rather than within individual districts. In many settings, the statewide approach would significantly reduce measurement differences between partisan and non-partisan maps. This improvement occurs because district estimates from a statewide ER/EI model of RPV can deviate from the truth for one of two reasons: statewide aggregation bias or substantial differences between the true statewide RPV and that of a specific district. The first source of bias is unaffected by mapmaker choices, while the second is likely minimal.¹⁴

In general, the statewide and within-district approaches represent two ends of a bias-variance dial. Statewide pooling offers little flexibility, but would be badly biased if there were significant heterogeneity in voting preferences across regions in a state. If, for example, political support by group first increases and then decreases with minority share, then a single EI model fit across the state’s full range of minority share would likely fail to approximate the true relationship. District-by-district inference has greater

¹⁴Adding and subtracting the true statewide RPV ($\bar{\Delta}$) from the state ER estimator ($E[\hat{\Delta}_{eco,s}|\bar{x}]$):

$$E[\hat{\Delta}_{eco,s}|\bar{x}] - \bar{\Delta}d = (E[\hat{\Delta}_{eco,s}|\bar{x}] - \bar{\Delta}) - (\bar{\Delta} - \bar{\Delta}_d) \quad (3)$$

The first term on the right-hand side of Equation 3, which captures statewide aggregation bias, is similar to Equation 2 just computed using all precincts in the state and centered on the statewide mean ($\bar{x}$). The second term reflects the difference between the true statewide RPV and district d ’s RPV. We conducted a series of simulation experiments to judge the relative significance of each source of bias. Those results, which above upon request, suggest the former source of bias accounts for nearly all of the typical measurement associated with the statewide approach.

flexibility, but also larger variance because each model is fit to a smaller set of precincts. The key practical challenge is that it is often difficult to judge which approach would be most appropriate precisely because group-specific vote shares are unobserved.

Fortunately, automated redistricting ensembles offer an alternative way to diagnose and account for modifiable-area-unit instability. Simulated ensembles can guide diagnostics by providing a sharp test for local aggregation bias. First, estimate within-district EI models for each simulated map in a representative ensemble. For each precinct in each plan, store the EI precinct predictions and the leave-one-out composition of the district it's placed in. Second, demean EI predictions and the racial composition of co-assigned precincts. Finally, estimate a within-precinct fixed-effects regression testing whether EI predictions track smoothly with the demographic composition of whichever precincts happen to be co-assigned. A significant slope would suggest the same precinct's estimate moves consistently as its district-mates' composition shifts, indicating persistent local aggregation bias.

In settings where researchers detect local bias, they can average precinct predictions from different representative maps to reduce noise. In the simplest case, researchers can store, for each precinct, the median predicted minority Democratic vote share and majority Democratic vote share across neutral simulated plans. Selecting the median prediction represents a relatively flat prior about the structure of bias; it assumes only that precinct's true β_i^b, β_i^w do not depend upon the composition of co-assigned precincts. More sophisticated selection procedures can incorporate additional spatial assumptions (e.g., that precinct parameters vary smoothly across the state or across district boundaries in the enacted plan). Likewise, researchers can also use redistricting ensembles to help partition states into regions (or sub-regions) within which racial groups may arguably be homogeneous.

Finally, researchers and practitioners can reduce model instability by incorporating additional information into the EI estimation model. For example, researchers can allow the prior mean of group-specific parameters (β^b, β^w) to depend explicitly on district composition by using the leave-one-out composition of each precincts co-assigned district-mates as a covariate in the EI likelihood model. The standard deviation of this quantity could provide a complementary measure of uncertainty. This measure could scale the variance of the group-specific EI parameters, shrinking precinct predictions more where district assignment is unstable (i.e., for precincts near political-geographic boundaries) and less where district assignment is highly stable (precincts embedded in homogeneous regions).

6 Conclusion

Researchers using ecological regression (ER) and ecological inference (EI) models are well aware of a range of common sources of model violations from contextual effects and spatial autocorrelation to model misspecification ([Greenland and Robins, 1994](#); [Achen and Shively, 1995](#); [Tam Cho, 1998](#); [Gelman et al., 2001](#)). Far fewer recognize that strategic redistricting may be an additional source of measurement bias. We demonstrate theoretically why a partisan gerrymander may choose to selectively aggregate precincts in ways that can induce correlations between racial composition and group-specific vote shares. Using simulations, we show that widely used ecological approaches for evaluating racially polarized voting (RPV) can perform particularly poorly gerrymandered maps, with the most severe distortions occurring when residential segregation is high.

Our findings suggest that differences in measurement error across mapmakers are especially large when ecological inference is conducted using models fit to precincts within (gerrymandered) districts. While differences across partisan and non-partisan

maps are smaller when ecological inference is conducted across all districts in our simulated state, it is difficult to judge which approach would be most appropriate in practice when group-specific vote shares are unobserved. One potential solution, which we would encourage researchers and practitioners to explore further, is to use simulation methods to conduct sensitivity analysis.

We would also encourage researchers to replicate and extend our framework – for example, by introducing precinct- or district-specific shocks to voter preferences by race – particularly in settings with high segregation, where ecological estimates are most likely to be misleading or even reverse the sign of estimated gaps between minority and majority voters turnout or candidate preferences. Whether using actual elections data or simulated data, our findings suggest that evaluating alternative assumptions about segregation, group-level preferences, and demographic covariates beyond race may provide valuable insights on the suitability of different ecological approaches. The same framework can also be used to evaluate more sophisticated ecological methods or survey-based approaches such as multilevel regression and post-stratification.

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A Formal Results

A.1 Preliminaries

Voter Preferences The prospective expected utility to a voter, i , with ideal point θ_i if the party D candidate is elected is

$$E[U_i(\text{party } D; \theta_i, \nu, \delta_i)] = -| -1 - \theta_i | + \nu + \mathbb{1}_{g_i=m} \delta_i$$

where $\mathbb{1}_{g_i=m}$ is an indicator function that takes value one if voter i belongs to the racial minority group, m , and zero otherwise. The voter's expected utility if a party R candidate is elected is

$$E[U_i(\text{party } R; \theta_i, \epsilon_i)] = -|1 - \theta_i|$$

Thus, members of the majority group support party R candidates if and only if $2\theta_i \geq \nu$,¹⁵ while minority voters support similar candidates if and only if $2\theta_i + \delta_i \geq \nu$. Let $p_R^g(\theta_i)$ denote the probability that a group g voter with ideal point θ_i supports party R . Let $\bar{x} \in (0, 1)$ denote the proportion of population belonging to the minority group.

While mapmakers cannot observe individual voters' preferences, they may have information about aggregate electoral support within a precinct or a district. Recall that $I_R(\theta)$ is the distribution of ideal points among party R 's core supporters and $I_D(\theta)$ is the corresponding distribution among out-party (party D) voters. Let $S_R^{gR}(\theta, \nu) = \int_0^1 p_R^g(\theta, \nu) dI_R(\theta)$ represent the cumulative support for party R from voters belonging to that party and group g . Similarly, let $S_R^{gD}(\theta, \nu) = \int_{-1}^0 p_R^g(\theta, \nu) dI_D(\theta)$ denote the party's cumulative support from cross-over (party D) group g voters.

¹⁵Note that $|1 - \theta_i| - |-1 - \theta_i| = -2\theta_i$

The aggregate support for party R before redistricting is:

$$\begin{aligned}
S_R(\pi, \nu) &= \bar{x} \left(\pi^m S_R^{mR} + (1 - \pi^m) S_R^{mD} \right) + (1 - \bar{x}) \left(\pi^w S_R^{wR} + (1 - \pi^w) S_R^{wD} \right) \\
&= \underbrace{\bar{x} \pi^m S_R^{mR} + (1 - \bar{x}) \pi^w S_R^{wR}}_{\text{Party R voter support}} + \underbrace{\bar{x} (1 - \pi^m) S_R^{mD} + (1 - \bar{x}) (1 - \pi^w) S_R^{wD}}_{\text{Cross-over support from party D voters}} \quad (4)
\end{aligned}$$

where π^g denotes the proportion of group g voters belonging to party R.

Legislative Redistricting Suppose that party R legislators hold a majority of legislative seats and therefore oversee redistricting. Party R's vote share on voting day will be $\pi + \Delta_\pi$ and its total support will be $S_R(\pi + \Delta_\pi, \nu)$.

Since S_R is weakly decreasing in ν , Party R's probability of winning must also decrease in ν . Let ν^* denote the cutoff value that solves $S_R(\pi + \Delta_\pi, \nu) = 0.5$. That is, the party R candidate wins at least 50% of votes in a district if and only if $\nu \leq \nu^*(\pi + \Delta_\pi)$. Given that ν is normally distributed, we can express the probability that party R wins seat j as:

$$\begin{aligned}
Pr(\text{R Wins}_j) &= Pr(S_R(\pi_j + \Delta_{\pi j}; \nu_j) \geq 1 - S_R(\pi_j + \Delta_{\pi j}; \nu_j)) \\
&= Pr(S_R(\pi_j + \Delta_{\pi j}; \nu_j) \geq \frac{1}{2}) \\
&= Pr(\nu \leq \nu^*(\pi_j + \Delta_{\pi j})) \\
&= \Phi(\nu^*(\pi_j + \Delta_{\pi j})) \quad (5)
\end{aligned}$$

In other words, the election outcome turns on whether the aggregate shock is favorable enough for party R candidates to push the party's support over 50%, given the normal vote (π) and the mapmaker's choice (π_Δ).

Let the function $M(\sigma)$ represent party R's ability to translate seat share into overall

legislative policy (where $\sigma \equiv \frac{1}{N} \sum_{j \in N} Pr(R \text{ Wins}_j)$ is the party R seat share after redistricting). To capture policy motivations, we assume that larger legislative majorities translate into greater policy benefits ($M' > 0$). To capture office-holding concerns, we assume the party derives some additional benefit from protecting party R incumbents. For simplicity, we assume that this benefit is the same for each of the party R incumbents and that it is increasing in the party's vote share in those districts.

The party's expected utility from implementing a new redistricting map, $\mathcal{M}^N$, is:

$$U_R(\mathcal{M}^N) = M(\sigma; \mathcal{M}^N) + \gamma \frac{1}{N_R} \sum_{j=1}^{N_R} Pr(R \text{ Wins}_j)$$

where $N_R \geq N/2$ is the number of seats controlled by party R , and $\gamma \in \mathbb{R}$ denotes the relative weight the party places on incumbency protection relative to policy motivations.

The party's equilibrium strategy will maximize:

$$\max_{\{\Delta_{\pi j}\}_{j=1}^N} M(\sigma; \mathcal{M}^N) + \gamma \frac{1}{N_R} \sum_{j=1}^{N_R} Pr(R \text{ Wins}_j)$$

such that $\sum_{j=1}^N \Delta_{\pi j} = 0$.

The first order conditions imply that:

$$\frac{1}{N} M'(\sigma; \mathcal{M}^N) \frac{\partial}{\partial \Delta_{\pi}} Pr(R \text{ Wins}_k) = \lambda, \forall k \in N \setminus N_R \quad (6)$$

$$(\gamma + \frac{1}{N} M'(\sigma; \mathcal{M}^N)) \frac{\partial}{\partial \Delta_{\pi}} Pr(R \text{ Wins}_j) = \lambda, \forall j \in N_R \quad (7)$$

where $\frac{\partial}{\partial \Delta_{\pi}} Pr(R \text{ Wins}_j)$ represents the change in party R 's win probability in district j after reassigning Δ_{π} to that district, and λ is the shadow price of the fixed-population constraint.

In words, Equation 6 states that, in the optimal plan, R mapmakers must be indiffer-

ent about which D -held district ought to receive a marginal increase in R voters. More generally, adding a friendly voter to any opposition seat must have the same effect on the party's policy utility (given by the product of the marginal increase in R 's win probability and the policy gain resulting from an additional seat). Otherwise, the party could re-allocate voters to increase its chances of electing one more co-partisan legislator without violating the population constraint. Equation 7 applies the same logic to seats currently controlled by party R , where the optimal allocation distributes voters to equalize their marginal impact on policy outcomes as well as incumbents' office-holding concerns.

A.2 Analysis

Strategic Redistricting without Racial Vote Polarization or Aggregation Bias We begin by reviewing conditions under which the anticipated electoral value of redistributing favorable voters to a district increases with the district's competitiveness. While our discussion will focus on districts currently controlled by party R (Equation 7), the optimal distribution of aligned voters to D districts follows a similar logic. The following lemma, adapted from [Gul and Pesendorfer \(2010\)](#), summarizes those conditions

Lemma 1 *The party R mapmaker's incentive to add more supportive voters to competitive party- R seats rather than safer ones is (a) increasing in partisan sorting i.e., the degree to which party affiliation predicts voting for party R , and (b) increasing in the extent to which partisan voters are concentrated at the ideological extremes rather than at the center.*

The first part of this lemma reflects the fact that reallocating partisan voters is more electorally valuable the more loyal those voters are to their respective parties. Partisan loyalties, which are captured by the partisan difference in aggregate support for party R

$(S_R/\partial\pi \geq 0)$, are built into model with the assumption that party D voters' ideal points are distributed to the left of their party R counterparts.

The second part of the lemma, which concerns the convexity of I_R and I_D , implies that safer districts will be less responsive to minor electoral shocks than competitive districts. As a result, assigning an additional aligned voter assigned to a safe party R district will yield smaller gains in electoral security than assigning that voter to a competitive district ($\frac{d^2 S_R/dv}{dS_R/dv} \geq 0$). Note that the mapmaker faces greater exposure to electoral shocks in competitive districts not only because those districts contain more out-party voters than party R strongholds. The disjoint support and convexity of I_R and I_D imply an additional cause for concern about competitive districts: larger electoral shocks attenuate the partisan difference in aggregate support more than smaller electoral shocks ($\frac{\partial}{\partial v} S_R^R - \frac{\partial}{\partial v} S_R^D \leq 0$). Allocating aligned voters to competitive districts becomes even more important if party R voters behave similarly to D after electoral shocks favoring the latter party.

Analysis with Vote Polarization and No Aggregation Bias Next, we evaluate whether the anticipated electoral benefit of strategic redistricting varies across districts with similar partisan compositions but different racial compositions. For now, we assume that the mapmaker is constitutionally forbidden from re-assigning voters to districts on the basis of group membership alone. We discuss this assumption in the context of strategic redistricting under aggregation bias below. To distinguish this case from the others, we use $\tilde{S}_R$ to denote party R s aggregate support under racial bloc voting but no aggregation bias.

Party R's aggregate support after redistricting is:

$$\begin{aligned}\tilde{S}_R(\pi + \Delta_\pi, \nu) = & \bar{x} \left((\pi^m + \Delta_\pi) \tilde{S}_R^{mR} + (1 - \pi^m - \Delta_\pi) \tilde{S}_R^{mD} \right) + \\ & + (1 - \bar{x}) \left((\pi^w + \Delta_\pi) \tilde{S}_R^{wR} + (1 - \pi^w - \Delta_\pi) \tilde{S}_R^{wD} \right)\end{aligned}\quad (8)$$

where $\tilde{S}_p^{mR} = 1 - I_p(\frac{\nu+\delta}{2})$ represents the party's cumulative support from party p minority voters and $\tilde{S}_p^{wR} = 1 - I_p(\frac{\nu}{2})$ represents the party's cumulative support from party p non-minority voters.

We begin by describing the relationship between minority share and party R support. Recall that partisan affiliation affects only the distribution of ideal points (θ) and bloc voting (δ_i) enters additively and identically for minority voters from either party. That is,

$$\begin{aligned}\tilde{S}_R^{mD} - \tilde{S}_R^{wD} &= \tilde{S}_R^{mR} - \tilde{S}_R^{wR}, \text{ or equivalently,} \\ \tilde{S}_R^{mD} - \tilde{S}_R^{mR} &= \tilde{S}_R^{wD} - \tilde{S}_R^{wR}\end{aligned}$$

To be sure, different groups may differ in their baseline support for party R (π^m may be greater than, smaller than or equal to π^w). But, the difference between typical party R and the party D voters' likelihood of casting a vote for party R is identical across racial groups: $\tilde{S}_R^{mD} - \tilde{S}_R^{mR} = \tilde{S}_R^{wD} - \tilde{S}_R^{wR}$. Differentiating Equation 8 highlights two reasons why the mapmaker might anticipate party R's probability of winning to be decreasing

in district minority share ($\frac{d}{d\bar{x}}\tilde{S}_R$)

$$\begin{aligned}
\frac{d}{d\bar{x}}\tilde{S}_R &= (\pi^m + \Delta_\pi)\tilde{S}_R^{mR} + (1 - \pi^m - \Delta_\pi)\tilde{S}_R^{mD} - (\pi^w + \Delta_\pi)\tilde{S}_R^{wR} - (1 - \pi^w - \Delta_\pi)\tilde{S}_R^{wD} \\
&= (\pi^m + \Delta_\pi)(\tilde{S}_R^{mR} - \tilde{S}_R^{mD}) - (\pi^w + \Delta_\pi)(\tilde{S}_R^{wR} - \tilde{S}_R^{wD}) + (\tilde{S}_R^{mD} - \tilde{S}_R^{wD}) \\
&= (\pi^m - \pi^w)(\tilde{S}_R^{wR} - \tilde{S}_R^{wD}) + (\tilde{S}_R^{mD} - \tilde{S}_R^{wD}) \\
&= \underbrace{(\pi^m - \pi^w) \left[I_D\left(\frac{\nu}{2}\right) - I_R\left(\frac{\nu}{2}\right) \right]}_{\text{racial differences in partisan sorting} \times \text{impact of partisan sorting}} + \underbrace{\left[I_D\left(\frac{\nu}{2}\right) - I_D\left(\frac{\nu + \delta}{2}\right) \right]}_{\text{impact of racial vote polarization on party } D \text{ voter's choices}}
\end{aligned}$$

Our main finding, summarized in *Proposition 1* below, concerns the optimal allocation of party R voters across seats with different racial compositions. Here, we focus on an electoral environment with idiosyncratic racial vote polarization ($\delta \geq 0$) but without any spatial correlation between minority share ($\bar{x}$) and partisan preferences by race. We relax this latter assumption in the following subsection.

Proposition 1. *Suppose the party R mapmaker has an incentive to add more supportive voters to competitive party- R seats rather than safer ones. Then, she also has an incentive to concentrate supportive voters in high minority share districts rather than low minority share districts. This latter incentive is (a) increasing in the degree of minority political cohesion, δ , and (b) increasing in the difference between the baseline proportion of minority voters and non-minority voters who support party D , $\pi^m - \pi^w$.*

Proposition 1 depends upon the conditions described in *Lemma 1*. The two specific effects operate through the same mechanisms. We focus on racial bloc voting, parameterized by $\delta > 0$, without imposing additional assumptions about other sources of racial vote polarization that may affect $\pi^m - \pi^w$.

A party R mapmaker who anticipates significant minority political cohesion will expect lower political support in districts with large minority populations ($\frac{d}{d\beta}\tilde{S}_R \leq 0$), even if minority and non-minority voters share similar partisan attachments ($\pi^m = \pi^w$). Thus, allocating Δ_π aligned voters to high minority share districts can dilute the impact of minority voters' political cohesion on expected electoral gains. This electoral benefit is larger in more competitive districts because of an interaction between political cohesion and voters' electoral response to valence shocks. In particular, (competitive) districts with higher minority shares will be more responsive to electoral shocks favoring party D than districts with the same partisan composition but lower minority shares. (This latter effect is captured by the last term in the cross-partial $\frac{d^2}{d\beta dv}\tilde{S}_R = (\pi^m - \pi^w)[I'_D(\frac{v}{2}) - I'_R(\frac{v}{2})] + [I'_D(\frac{v}{2}) - I'_D(\frac{v+\delta}{2})]$)

Analysis with Vote Polarization and Aggregation Bias Our goal in this subsection is to derive implications of strategic redistricting for measurement performance. To that end, we introduce a specific form of aggregation bias into the voter model. We then describe how mapmakers' partisan incentives may affect the degree of aggregation bias relative to a non-partisan map-making process.

First, we introduce an additional source of local variation in vote choice that may be correlated with local demographic patterns. In particular, we assume each voter in a precinct p derives an idiosyncratic benefit (ϵ_p) such that voters with $\epsilon_p > 0$ are more likely to support party R candidates while voters with $\epsilon_p < 0$ prefer party D candidates. ϵ_p is independently and uniformly distributed over the interval $[-\bar{\epsilon} + B(\bar{x}_p), \bar{\epsilon} + B(\bar{x}_p)]$. The parameter $\bar{\epsilon} > 0$ represents the aggregate preference heterogeneity in the electorate stemming from factors beyond ideological preference or racial bloc voting. The function $B(\bar{x})$ captures the relationship between minority share and aggregate support for party R , and the slope $\frac{dB}{d\bar{x}} \neq 0$ reflects a form of ecological bias called intercept bias (see Section

2.2).

While a voter's expected utility if a party D candidate is elected remains unchanged:

$$E[U_i(\text{party } D; \theta_i, v)] = -| -1 - \theta_i | + v + \mathbb{1}_{g_i=m} \delta_i$$

their corresponding expected utility if the party R candidate is elected becomes:

$$E[U_i(\text{party } R; \theta_i, \epsilon_i)] = -|1 - \theta_i| + \epsilon_p$$

Moving forward, we use $\hat{p}_R$ and $\hat{S}_R$ to denote an individual voter's party R support and total party R support for this case. Because ϵ_p is uniformly distributed, each non-minority (white) voter's probability of casting a party R vote is

$$\begin{aligned} \hat{p}_R^w &= \Pr(\epsilon_p \geq v - 2\theta_i) = \Pr\left(\frac{\bar{\epsilon} + B(\bar{x}) - (v - 2\theta_i)}{2\bar{\epsilon}}\right) \\ &= \Pr\left(\theta \geq \frac{v - \bar{\epsilon} - B(\bar{x})}{2}\right) \end{aligned}$$

Let $\theta_0^w \equiv \frac{v - \bar{\epsilon} - B(\bar{x})}{2}$ denote the ideal point cutoff such that those white voters with $\theta \geq \theta_0^w$ cast votes for party R and white voters with $\theta < \theta_0^w$ do not. Minority bloc voting raises the corresponding threshold for minority voters, which we denote by $\theta_0^m = \theta_0^w + \frac{\delta}{2}$. We will refer to the share of group- g voters (across both parties) with ideal points to the right of the threshold by $1 - I^g(\theta) = 1 - \pi^g I_R(\theta_0^g) + (1 - \pi^g) I_D(\theta_0^g)$.

Prior to redistricting, the party's aggregate support from group g voters will be

$$\hat{S}_R^g(\pi^g, v) = (\pi^g) \int_{\theta_0^g}^1 \hat{p}_R^g dI_R(\theta) + (1 - \pi^g) \int_{\theta_0^g}^1 \hat{p}_R^g dI_D(\theta)$$

The Party R candidate wins if their aggregate support after redistricting exceeds 50%

of the electorate

$$\hat{S}_R(\pi + \Delta_\pi, \nu) = \bar{x} \hat{S}_R^m(\pi^g + \Delta_\pi, \nu) + (1 - \bar{x}) \hat{S}_R^w(\pi^g + \Delta_\pi, \nu) \geq 0.5$$

Let $\Delta \equiv (\hat{S}_R^m - \hat{S}_R^w)$ represent the true degree of vote polarization. In practice, we typically only observe precinct minority-share and vote share $(\bar{x}_p, \hat{S}_{Rp})$. Next, we show that regression of vote share on minority share cannot distinguish between two observationally equivalent explanations for why high-minority precincts show different R support: (i) minority voters themselves may be relatively D or R -friendly or (ii) high-minority precincts simply happen to draw a more D or R -favorable aggregate political climate.

Differentiating precinct-level party vote share with respect to precinct minority share

$$\begin{aligned} \frac{\partial}{\partial \bar{x}} \hat{S}_R &= \hat{S}_R^m + \bar{x} \frac{\partial}{\partial \bar{x}} \hat{S}_R^m - \hat{S}_R^w + (1 - \bar{x}) \frac{\partial}{\partial \bar{x}} \hat{S}_R^w \\ \frac{\partial}{\partial \bar{x}} \hat{S}_R - \Delta &= \frac{\partial}{\partial \bar{x}} \hat{S}_R^w + \bar{x} \frac{\partial}{\partial \bar{x}} \Delta \\ &= \frac{B'(\bar{x})}{2\bar{\epsilon}} [\bar{x}(1 - I^m(\theta)) + (1 - \bar{x})(1 - I^w(\theta))] \end{aligned} \quad (9)$$

Suppose, for example, that high minority share precincts carry systematically stronger party R voters than low minority share precincts (i.e. $B'(\bar{x}) \geq 0$). Equation 9 suggests that the ecological regression regression slope will tend to exaggerate the degree of vote polarization (Δ); further, this deviation is proportional to the share of party R voters and is increasing in the importance of contextual effects ($B'(\bar{x}) \geq 0$) relative to all other sources of non-ideological preference heterogeneity ($\bar{\epsilon}$).

The magnitude of this distortion within any district will depend upon both political geography and mapmaker choices. The key implication of *Proposition 1* is that partisan considerations will favor redistricting maps that allocate politically aligned precincts with low minority shares to (competitive) districts with higher minority shares. We now

show that this form of gerrymandering amplifies aggregation bias in whichever direction $B'(\bar{x}) \geq 0$ already points.

Before doing that, it is important to notice that a shift in partisan composition (Δ_π) may or may not be applied symmetrically to both racial groups. If Δ_π is applied symmetrically to both groups, so that the target district's minority share remains unchanged, then Δ_π only affects aggregation bias via a shift in the partisan compositions $(\bar{x}(1 - I^m(\theta)) + (1 - \bar{x})(1 - I^w(\theta)))$. In particular, shifting partisan preferences from the pair (π^m, π^w) to the pair $(\pi^m + \Delta_\pi, \pi^w + \Delta_\pi)$ shifts the right hand-side of Equation 9 by $\frac{B'(\bar{x})}{2\bar{\epsilon}}[\bar{x}(\Gamma^m(\theta) + (1 - \bar{x})\Gamma^w(\theta))]$ where $\Gamma^g = 1 - I_R(\theta_0^g) - I_D(\theta_0^g)$. Because $\Gamma^g \geq 0$ for any θ , shifting Δ_π voters amplifies or attenuates aggregation bias while leaving the sign of aggregation bias unaffected.

Now suppose that mapmakers are literally moving population from a low-minority to a high-minority share district. In this case, a Δ_π shift may mechanically dilute minority share in the latter district, which may affect both terms on the right hand side of Equation 9: placing the district at a lower level of the $B(\bar{x})$ curve and reweighting the population toward the preference of white voters.

To generalize this argument, we derive the expected effect of redistricting on measurement performance in section A.3.3. Specifically, we consider a precinct, p , whose minority share and opposition (party D) vote share are sufficiently low that adding it to a nearby district reduces both quantities for that district. We can describe the impact of the precinct's reassignment on ecological estimates of RPV. Specifically, we can show that redistricting can either increase or reduce ecological bias, amplifying bias when estimates are already upward-biased and attenuating it otherwise.

A.3 Proofs

A.3.1 Lemma 1.

To fix ideas, consider strategic redistricting in an environment without idiosyncratic racial vote polarization ($\delta = 0$). In this case, a voter i supports party R if and only if $\theta_i \geq \frac{\nu}{2}$. Party R's aggregate support can be decomposed into the mass of party R voters and the mass of party D voters to the right of this cutoff. Further, we can simplify Equation 4 by defining

$$\begin{aligned} S_R^R &= S_R^{wR} = S_R^{mR} = 1 - I_R\left(\frac{\nu}{2}\right) \\ S_D^R &= S_D^{wR} = S_D^{mR} = 1 - I_D\left(\frac{\nu}{2}\right). \end{aligned}$$

For expositional brevity, we omit explicit references to $\delta = 0$ throughout the remainder of this case. Noting that $\pi \equiv \beta\pi^m + (1 - \beta)\pi^w$, we can express party R's total support on voting day by $S_R(\nu, \pi + \Delta_\pi) = (\pi + \Delta_\pi)(1 - I_R(\frac{\nu}{2})) + (1 - \pi - \Delta_\pi)(1 - I_D(\frac{\nu}{2}))$. Begin by differentiating the implicit cutoff condition ($S_R(\pi + \Delta_\pi, \nu^*) = \frac{1}{2}$) as follows:

$$\begin{aligned} \frac{d}{d\Delta_\pi} S_R(\nu^*, \pi + \Delta_\pi) &= 0 \\ \frac{\partial S_R}{\partial \nu} \frac{d\nu^*}{d\Delta_\pi} + \frac{\partial S_R}{\partial \Delta_\pi} &= 0 \\ \frac{d\nu^*}{d\Delta_\pi} &= -\frac{\partial S_R / \partial \Delta_\pi}{\partial S_R / \partial \nu} \end{aligned} \tag{10}$$

By the chain rule, $\frac{\partial}{\partial \Delta_\pi} Pr(\text{R Wins}) = \phi(\nu^*) \frac{d\nu^*}{d\Delta_\pi} \geq 0$. Next, differentiate this term with

respect to party R's normal vote share. Noting that $\phi'(\nu^*) = \nu^* \phi(\nu^*)$ and that $\frac{d\nu^*}{d\pi} = \frac{d\nu^*}{d\Delta_\pi}$

$$\begin{aligned} \frac{\partial}{\partial \pi \partial \Delta_\pi} \text{Pr(R Wins)} &= \phi'(\nu^*) \frac{d\nu^*}{d\pi} \frac{d\nu^*}{d\Delta_\pi} + \phi(\nu^*) \frac{d\nu^*}{d\pi d\Delta_\pi} \\ &= \phi(\nu^*) \left(\frac{\phi'(\nu^*)}{\phi(\nu^*)} \left(\frac{d\nu^*}{d\pi} \right)^2 + \frac{d\nu^*}{d\pi d\Delta_\pi} \right) \end{aligned} \quad (11)$$

To solve for $\frac{d}{d\pi d\Delta_\pi} \nu^*$, differentiate Equation 10 by party R's normal vote share:

$$\begin{aligned} \frac{d}{d\pi} \left(\frac{\partial S_R}{\partial \nu} \frac{d\nu^*}{d\Delta_\pi} + \frac{\partial S_R}{\partial \Delta_\pi} \right) &= 0 \\ \frac{d}{d\pi} \left(\frac{\partial S_R}{\partial \nu} \frac{d\nu^*}{d\Delta_\pi} \right) + \frac{d}{d\pi} \left(\frac{\partial S_R}{\partial \Delta_\pi} \right) &= 0 \\ \left(\frac{\partial^2 S_R}{\partial \nu} \frac{d\nu^*}{d\pi} \frac{d\nu^*}{d\Delta_\pi} + \frac{\partial^2 S_R}{\partial \pi \partial \nu} \frac{d\nu^*}{d\Delta_\pi} + \frac{\partial S_R}{\partial \nu} \frac{d}{d\pi d\Delta_\pi} \nu^* \right) + \left(\frac{\partial^2 S_R}{\partial \nu \partial \Delta_\pi} \frac{d\nu^*}{d\pi} + \frac{\partial^2 S_R}{\partial \pi \partial \Delta_\pi} \right) &= 0 \\ \frac{d}{d\pi d\Delta_\pi} \nu^* &= - \frac{(\partial^2 S_R / \partial \nu) \left(\frac{d\nu^*}{d\pi} \right)^2 + 2(\partial^2 S_R / \partial \Delta_\pi \partial \nu) \frac{d\nu^*}{d\pi} + (\partial^2 S_R / \partial \pi \partial \Delta_\pi)}{\partial S_R / \partial \nu} \end{aligned}$$

Noting that $\frac{\partial^2 S_R}{\partial \pi \partial \nu} = \frac{\partial^2 S_R}{\partial \Delta_\pi \partial \nu} = \frac{\partial}{\partial \nu} S_R^R - \frac{\partial}{\partial \nu} S_R^D$, and that $\partial^2 S_R / \partial \pi \partial \Delta_\pi = 0$

Equation 12 summarizes the conditions described in Lemma 1. The mapmaker has a greater incentive to assign supportive voters to competitive party-R seats rather than safer ones ($\frac{\partial \text{Pr(R Wins)}}{\partial \pi \partial \Delta_\pi} \leq 0$) if and only if

$$\begin{aligned} \frac{\phi'(\nu^*)}{\phi(\nu^*)} \left(\frac{d\nu^*}{d\pi} \right)^2 - \frac{(\partial^2 S_R / \partial \nu) \left(\frac{d\nu^*}{d\pi} \right)^2 + 2 \left(\frac{\partial}{\partial \nu} S_R^R - \frac{\partial}{\partial \nu} S_R^D \right) \frac{d\nu^*}{d\pi}}{\partial S_R / \partial \nu} &\leq 0 \\ \frac{\phi'(\nu^*)}{\phi(\nu^*)} \left(\frac{d\nu^*}{d\pi} \right)^2 &\leq \frac{\partial^2 S_R / \partial \nu}{\partial S_R / \partial \nu} \left(\frac{d\nu^*}{d\pi} \right)^2 + 2 \frac{\left(\frac{\partial}{\partial \nu} S_R^R - \frac{\partial}{\partial \nu} S_R^D \right) \frac{d\nu^*}{d\pi}}{\partial S_R / \partial \nu} \\ \frac{\phi'(\nu^*)}{\phi(\nu^*)} &\leq \frac{\partial^2 S_R / \partial \nu}{\partial S_R / \partial \nu} + 2 \frac{\left(\frac{\partial}{\partial \nu} S_R^R - \frac{\partial}{\partial \nu} S_R^D \right)}{(\partial S_R / \partial \nu) \frac{d\nu^*}{d\pi}} \\ \frac{\phi'(\nu^*)}{\phi(\nu^*)} &\leq \frac{\partial^2 S_R / \partial \nu}{\partial S_R / \partial \nu} - 2 \frac{\left(\frac{\partial}{\partial \nu} S_R^R - \frac{\partial}{\partial \nu} S_R^D \right)}{\partial S_R / \partial \pi}, \end{aligned} \quad (12)$$

where $\frac{\partial}{\partial \nu} S_R^R - \frac{\partial}{\partial \nu} S_R^D = I'_D(\frac{\nu}{2}) - I'_R(\frac{\nu}{2}) \leq 0$;

$$\begin{aligned}\frac{d^2 S_R / dv}{dS_R / dv} &= \frac{-\frac{1}{4}(\pi + \Delta_\pi) I_R''(\frac{v}{2}) - \frac{1}{4}(1 - \pi - \Delta_\pi) I_D''(\frac{v}{2})}{-\frac{1}{2}(\pi + \Delta_\pi) I_R'(\frac{v}{2}) - \frac{1}{2}(1 - \pi - \Delta_\pi) I_D'(\frac{v}{2})} \\ &= \frac{1(\pi + \Delta_\pi) I_R''(\frac{v}{2}) + (1 - \pi - \Delta_\pi) I_D''(\frac{v}{2})}{2(\pi + \Delta_\pi) I_R'(\frac{v}{2}) + (1 - \pi - \Delta_\pi) I_D'(\frac{v}{2})} \geq 0;\end{aligned}$$

and, by assumption, the partisan difference in aggregate support for party R is strictly positive ¹⁶ i.e., $\partial S_R / \partial \pi = S_R^R - S_R^D = I_D(\frac{v}{2}) - I_R(\frac{v}{2}) \geq 0$

A.3.2 Proposition 1.

From the implicit function theorem,

$$\frac{d\Delta_\pi^*}{d\beta} = -\frac{(\gamma + \frac{1}{N}M') \frac{\partial^2}{\partial \beta \partial \Delta_\pi} Pr(\text{R Wins})}{(\gamma + \frac{1}{N}M') \frac{\partial^2}{\partial \Delta_\pi} Pr(\text{R Wins})} = -\frac{\frac{\partial^2}{\partial \beta \partial \Delta_\pi} Pr(\text{R Wins})}{\frac{\partial^2}{\partial \Delta_\pi} Pr(\text{R Wins})} \quad (13)$$

The denominator of this expression is described by the conditions identified in the preceding subsection (see Equation 12). The numerator is:

$$\begin{aligned}\frac{\partial^2}{\partial \beta \partial \Delta_\pi} Pr(\text{R Wins}) &= \phi'(v^*) \frac{dv^*}{d\beta} \frac{dv^*}{d\Delta_\pi} + \phi(v^*) \frac{dv^*}{d\beta d\Delta_\pi} \\ &= \phi(v^*) \left(\frac{\phi'(v^*)}{\phi(v^*)} \frac{\partial v^*}{\partial \beta} \frac{\partial v^*}{\partial \Delta_\pi} + \frac{\partial^2 v^*}{\partial \beta \partial \Delta_\pi} \right)\end{aligned}$$

¹⁶Because party D voters' ideal points are distributed to the left of their party R counterparts: $\frac{\partial S_R}{\partial \Delta_\pi} = S_R^R - S_R^D = 1 - I_R(\frac{v}{2}) - (1 - I_D(\frac{v}{2})) = I_D(\frac{v}{2}) - I_R(\frac{v}{2}) \geq 0$

To evaluate the cross-partial $\frac{dv^*}{d\beta d\Delta_\pi}$, differentiate Equation 11:

$$\begin{aligned} \frac{d}{d\beta} \left(\frac{\partial \tilde{S}_R}{\partial v} \frac{dv^*}{d\Delta_\pi} + \frac{\partial \tilde{S}_R}{\partial \Delta_\pi} \right) &= 0 \\ \frac{d}{d\beta} \left(\frac{\partial \tilde{S}_R}{\partial v} \frac{dv^*}{d\Delta_\pi} \right) + \frac{d}{d\beta} \left(\frac{\partial \tilde{S}_R}{\partial \Delta_\pi} \right) &= 0 \\ \left(\frac{\partial^2 \tilde{S}_R}{\partial v} \frac{dv^*}{d\beta} \frac{dv^*}{d\Delta_\pi} + \frac{\partial^2 \tilde{S}_R}{\partial \beta \partial v} \frac{dv^*}{d\Delta_\pi} + \frac{\partial \tilde{S}_R}{\partial v} \frac{d}{d\beta d\Delta_\pi} v^* \right) + \left(\frac{\partial^2 \tilde{S}_R}{\partial v \partial \Delta_\pi} \frac{dv^*}{d\beta} + \frac{\partial^2 \tilde{S}_R}{\partial \beta \partial \Delta_\pi} \right) &= 0 \\ \frac{d}{d\beta d\Delta_\pi} v^* &= - \frac{(\partial^2 \tilde{S}_R / \partial v) \frac{dv^*}{d\beta} \frac{dv^*}{d\Delta_\pi} + \frac{\partial^2 \tilde{S}_R}{\partial \beta \partial v} \frac{dv^*}{d\Delta_\pi} + \frac{\partial^2 \tilde{S}_R}{\partial v \partial \Delta_\pi} \frac{dv^*}{d\beta} + \frac{\partial^2 \tilde{S}_R}{\partial \beta \partial \Delta_\pi}}{\partial \tilde{S}_R / \partial v} \end{aligned}$$

where $\frac{d^2}{d\beta d\Delta_\pi} \tilde{S}_R = 0$. Finally, note that $\frac{\partial^2}{\partial \beta \partial \Delta_\pi} Pr(\text{R Wins}) \geq 0$ if and only if

$$\begin{aligned} \frac{\phi'(v^*)}{\phi(v^*)} \frac{\partial v^*}{\partial \beta} \frac{\partial v^*}{\partial \Delta_\pi} + \frac{\partial^2 v^*}{\partial \beta \partial \Delta_\pi} &\geq 0 \\ \frac{\phi'(v^*)}{\phi(v^*)} &\geq \frac{d^2 \tilde{S}_R / dv}{d\tilde{S}_R / dv} + \frac{(d^2 \tilde{S}_R / d\beta dv)}{(d\tilde{S}_R / dv) \frac{\partial v^*}{\partial \beta}} + \frac{(d^2 \tilde{S}_R / dv d\Delta_\pi)}{(d\tilde{S}_R / dv) \frac{\partial v^*}{\partial \Delta_\pi}} + \frac{d^2 \tilde{S}_R / d\beta d\Delta_\pi}{\frac{\partial v^*}{\partial \beta} \frac{\partial v^*}{\partial \Delta_\pi}} \\ \frac{\phi'(v^*)}{\phi(v^*)} &\geq \frac{d^2 \tilde{S}_R / dv}{d\tilde{S}_R / dv} - \frac{(d^2 \tilde{S}_R / d\beta dv)}{(d\tilde{S}_R / d\beta)} - \frac{(d^2 \tilde{S}_R / dv d\Delta_\pi)}{(d\tilde{S}_R / d\Delta_\pi)} \end{aligned}$$

Suppose $\frac{\partial^2}{\partial \Delta_\pi} Pr(\text{R Wins}) = \frac{\partial^2}{\partial \pi \partial \Delta_\pi} Pr(\text{R Wins}) \leq 0$ (see Equation 12). Thus, Equation 13 implies that $\frac{d\Delta_\pi^*}{d\beta} \geq 0$ if and only if

$$\frac{\phi'(v^*)}{\phi(v^*)} \leq \frac{d^2 \tilde{S}_R / dv}{d\tilde{S}_R / dv} - \frac{(d^2 \tilde{S}_R / d\beta dv)}{(d\tilde{S}_R / d\beta)} - \frac{(d^2 \tilde{S}_R / dv d\Delta_\pi)}{(d\tilde{S}_R / d\Delta_\pi)} \quad (14)$$

Note that if $\pi^m \leq \pi^w$, then $\frac{d\tilde{S}_R}{d\beta} \leq 0$,

$$\frac{d^2}{d\beta dv} \tilde{S}_R = (\pi^m - \pi^w) \left[I'_D\left(\frac{v}{2}\right) - I'_R\left(\frac{v}{2}\right) \right] + \left[I'_D\left(\frac{v}{2}\right) - I'_D\left(\frac{v+\delta}{2}\right) \right] \geq 0$$

In short, if $\pi^m \leq \pi^w$ and the party R mapmaker has an incentive to assign more supportive voters to competitive seats rather than safer ones, then she also has an incentive to add more supportive voters to higher-minority share seats rather than lower minority

share seats.

A.3.3 Deriving implications of strategic redistricting for measurement.

We observe a set of precincts, indexed by $j \in S_d$, in district d . Each precinct includes voters $i=1, \dots, n_j$. Those voters may belong to a racial minority group, as indicated by the dummy variable $x_i \in \{0, 1\}$, and must cast their vote for either party D or party R candidates; the outcome $\text{Vote } D_i \in \{0, 1\}$ is an indicator for party D voters.

To avoid confusion, we use numeric subscripts 0 and 1 (e.g. n_0 and n_1) to denote district parameters (e.g. population) before and after the precinct's reassignment, and the subscript p (e.g. n_p) to denote the additional precinct. The following table summarizes key district and precinct parameters.

Object	Interpretation
n	Total population
n^g	Number of group g voters
β^g	Proportion of opposition party voters in group g
$\bar{x} \equiv \frac{n^m}{n^w + n^m}$	Minority share
$\bar{v} \equiv \beta^w(1 - \bar{x}) + \beta^m \bar{x}$	Opposition party vote share
$\Delta \equiv \beta^m - \beta^w$	Degree of racial vote polarization

First, noting that $\sum_{i=1}^m n_i \bar{x}_i = n \bar{x}_0 = \sum_{i=1}^m n_i \bar{x}_0$, we have $\sum_{i=1}^m n_i \bar{v}_0 (\bar{x}_i - \bar{x}_0) = 0$.

$$\begin{aligned} \bar{\Delta}^{eco} &= \frac{\sum_{i=1}^m n_i (\bar{x}_i - \bar{x}_1) (\bar{v}_i - \bar{v}_1)}{\sum_{i=1}^m n_i (\bar{x}_i - \bar{x}_1)^2} \\ &= \frac{\sum_{i=1}^m n_i \bar{v}_i (\bar{x}_i - \bar{x}_1)}{\sum_{i=1}^m n_i (\bar{x}_i - \bar{x}_1)^2} \end{aligned}$$

Second, noting that $\bar{x}_1 - \bar{x}_p = \frac{n}{n+1}(\bar{x}_0 - \bar{x}_p)$, the coefficient in Equation 1 after the

precinct's reassignment will be

$$\begin{aligned}
E[\tilde{\Delta}_1^{eco}|\bar{\mathbf{x}}_1] &= \frac{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)(E[\bar{v}_i|\bar{x}_i] - \bar{v}_1)}{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)^2} \\
&= \frac{(\bar{x}_p - \bar{x}_1)(\bar{v}_p - \bar{v}_1) + \sum_{i=1}^M n_i(E[\bar{v}_i|\bar{x}_i] - \bar{v}_1)(\bar{x}_i - \bar{x}_1)}{(\bar{x}_p - \bar{x}_1)(\bar{x}_p - \bar{x}_1) + \sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_1)^2} \\
&= \frac{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)(\bar{v}_p - \bar{v}_0) + \sum_{i=1}^M n_i E[\bar{v}_i|\bar{x}_i](\bar{x}_i - \bar{x}_0)}{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)^2 + \sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)^2}
\end{aligned}$$

Therefore, the precinct's reassignment changes the ER coefficient by

$$\begin{aligned}
E[\tilde{\Delta}_1^{eco}|\bar{\mathbf{x}}_1] - E[\tilde{\Delta}_0^{eco}|\bar{\mathbf{x}}_0] &= \frac{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)E[\bar{v}_i|\bar{x}_i]}{\sum_{i=1}^{M+1} n_i(\bar{x}_i - \bar{x}_1)^2} - \frac{\sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)E[\bar{v}_i|\bar{x}_i]}{\sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)^2} \\
&= \frac{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)(\bar{v}_p - \bar{v}_0) + \sum_{i=1}^M n_i E[\bar{v}_i|\bar{x}_i](\bar{x}_i - \bar{x}_0)}{(\frac{n}{n+1})^2(\bar{x}_p - \bar{x}_0)^2 + \sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)^2} - \frac{\sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)E[\bar{v}_i|\bar{x}_i]}{\sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)^2} \\
&= h(\bar{x}_p - \bar{x}_0)[(\bar{v}_p - \bar{v}_0) - (\bar{x}_p - \bar{x}_0)E[\tilde{\Delta}_0^{eco}|\bar{\mathbf{x}}_0]] \\
&= h(\bar{x}_0 - \bar{x}_p)[(\bar{v}_0 - \bar{v}_p) - (\bar{x}_0 - \bar{x}_p)E[\tilde{\Delta}_0^{eco}|\bar{\mathbf{x}}_0]]
\end{aligned}$$

where $h \equiv \frac{c^2}{c^2(\bar{x}_p - \bar{x}_0)^2 + \sum_{i=1}^M n_i(\bar{x}_i - \bar{x}_0)^2}$ and $c \equiv \frac{n}{n+1}$.

By definition, $h(\bar{x}_0 - \bar{x}_p) \geq 0$, and we can rewrite the term $\bar{v}_p - \bar{v}_0$ as follows

$$\begin{aligned}
\bar{v}_0 - \bar{v}_p &= \beta_x^w(1 - \bar{x}_x) + \beta_0^m \bar{x}_0 - \beta_p^w(1 - \bar{x}_p) - \beta_p^m \bar{x}_p \\
&= \beta_0^w - \beta_p^w + \bar{x}_0(\beta_0^m - \beta_0^w) - \bar{x}_p(\beta_p^m - \beta_p^w) \\
&= \beta_0^w - \beta_p^w + \bar{x}_0\Delta_0 - \bar{x}_p\Delta_p \\
&= \beta_0^w - \beta_p^w + \bar{x}_0\Delta_0 - \bar{x}_p\Delta_0 + \bar{x}_p\Delta_0 - \bar{x}_p\Delta_p \\
&= \kappa + (\bar{x}_0 - \bar{x}_p)\Delta_0
\end{aligned}$$

where $\kappa \equiv \beta_0^w - \beta_p^w + \bar{x}_p(\Delta_0 - \Delta_p)$

Therefore, we can show that the ecological estimate increases after the precinct's reassignment ($E[\tilde{\Delta}_1^{eco}|\bar{\mathbf{x}}_1] \geq E[\tilde{\Delta}_0^{eco}|\bar{\mathbf{x}}_0]$) if and only if

$$\begin{aligned}\frac{\bar{v}_0 - \bar{v}_p}{\bar{x}_0 - \bar{x}_p} &\geq E[\tilde{\Delta}_0^{eco}|\bar{\mathbf{x}}_0] \\ \frac{\kappa}{\bar{x}_0 - \bar{x}_p} &\geq E[\tilde{\Delta}_0^{eco}|\bar{\mathbf{x}}_0] - \Delta_0\end{aligned}\tag{15}$$

By contrast, true racial vote polarization increases ($\Delta_1 \geq \Delta_0$) if and only if

$$\begin{aligned}\frac{\beta_0^m n_0^m + \beta_p^m n_p^m}{n_0^m + n_p^m} - \frac{\beta_0^w n_0^w + \beta_p^w n_p^w}{n_0^w + n_p^w} &\geq \frac{\beta_0^m n_0^m}{n_0^m} - \frac{\beta_0^w n_0^w}{n_0^w} \\ \frac{\beta_0^w n_0^w}{n_0^w} - \frac{\beta_0^w n_0^w + \beta_p^w n_p^w}{n_0^w + n_p^w} &\geq \frac{\beta_0^m n_0^m}{n_0^m} - \frac{\beta_0^m n_0^m + \beta_p^m n_p^m}{n_0^m + n_p^m} \\ n_0^w n_p^w \frac{\beta_0^w - \beta_p^w}{n_0^w (n_0^w + n_p^w)} &\geq n_0^m n_p^m \frac{\beta_0^m - \beta_p^m}{n_0^m (n_0^m + n_p^m)} \\ \frac{\beta_0^w - \beta_p^w}{\beta_0^m - \beta_p^m} &\geq \frac{n_p^m}{(n_0^m + n_p^m)} \frac{(n_0^w + n_p^w)}{n_p^w} = \frac{\bar{x}_p(1 - \bar{x}_1)}{(1 - \bar{x}_p)\bar{x}_1} \\ \beta_0^w - \beta_p^w &\geq (\beta_0^m - \beta_p^m) \left(\frac{\bar{x}_p(1 - \bar{x}_1)}{(1 - \bar{x}_p)\bar{x}_1} \right) \\ \beta_0^w - \beta_p^w &\geq (\beta_0^w + \Delta_0 - \beta_p^w - \Delta_p) \left(\frac{\bar{x}_p(1 - \bar{x}_1)}{(1 - \bar{x}_p)\bar{x}_1} \right) \\ (\beta_0^w - \beta_p^w) \frac{(1 - \bar{x}_p)\bar{x}_1}{(1 - \bar{x}_1)} &\geq (\beta_0^w + \Delta_0 - \beta_p^w - \Delta_p) \bar{x}_p \\ (\beta_0^w - \beta_p^w) \frac{(1 - \bar{x}_p)\bar{x}_1}{(1 - \bar{x}_1)} + (\beta_0^w - \beta_p^w)(1 - \bar{x}_p) &\geq (\beta_0^w - \beta_p^w) + (\Delta_0 - \Delta_p) \bar{x}_p \\ (\beta_0^w - \beta_p^w) \frac{(1 - \bar{x}_p)}{\bar{x}_1(1 - \bar{x}_1)} &\geq \kappa\end{aligned}$$

Dividing by $\bar{x}_0 - \bar{x}_p$, racial vote polarization increases ($\Delta_1 \geq \Delta_0$) if and only if

$$\frac{\kappa}{\bar{x}_0 - \bar{x}_p} \leq (\beta_0^w - \beta_p^w) \frac{(1 - \bar{x}_p)}{(\bar{x}_0 - \bar{x}_p)(\bar{x}_1)(1 - \bar{x}_1)}\tag{16}$$

Putting this together, the precinct's reassignment will produce upward sign bias if the condition in Equation 15 is satisfied but the one in Equation 16 is not. The likelihood that this occurs is

$$Pr\left(\frac{\kappa}{\bar{x}_0 - \bar{x}_p} \geq \max \left\{ E[\bar{\Delta}_0^{eco} | \bar{\mathbf{x}}_0] - \Delta_0, (\beta_0^w - \beta_p^w) \frac{(1 - \bar{x}_p)}{\bar{x}_1(\bar{x}_0 - \bar{x}_p)(1 - \bar{x}_1)} \right\} \right)$$

Likewise, the likelihood that the precinct's reassignment will produce downward-biased ecological estimates is

$$Pr\left(\frac{\kappa}{\bar{x}_0 - \bar{x}_p} \leq \min \left\{ E[\bar{\Delta}_0^{eco} | \bar{\mathbf{x}}_0] - \Delta_0, (\beta_0^w - \beta_p^w) \frac{(1 - \bar{x}_p)}{\bar{x}_1(\bar{x}_0 - \bar{x}_p)(1 - \bar{x}_1)} \right\} \right)$$

B Measurement Results

B.1 Decomposition of bias

In this paper, we are primarily interested in the difference between district minority support rate and district majority support rate :

$$\bar{\Delta}_d \equiv \underbrace{\frac{\sum_{j \in S_d} n_j \bar{x}_j \beta_j^b}{n_d \bar{x}_d}}_{\text{Minority Party } D \text{ Rate}} - \underbrace{\frac{\sum_{j \in S_d} n_j (1 - \bar{x}_j) \beta_j^w}{n_d (1 - \bar{x}_d)}}_{\text{Majority Party } D \text{ Rate}},$$

One conceptual advantages of this estimand over the average (population-weighted) difference in vote shares by race ($\bar{\Delta}_{n,d} \equiv \frac{1}{n_d} \sum_{j \in S_d} n_j \Delta_j$) is that $\bar{\Delta}_d$ clarifies the role of racial residential segregation on measurement performance. Specifically, we can decompose aggregation bias into

$$\begin{aligned} E[\bar{\Delta}_{eco,d} | \bar{x}_d] - \bar{\Delta}_d &= (1 - \rho_{\bar{x}_d}) \frac{Cov_n(x, \beta^w)}{Var_n(x)} \\ &+ (1 - \frac{1 - \bar{x}_d}{\bar{x}_d} \rho_{\bar{x}_d}) \frac{Cov_n(x, \Delta)}{Var_n(x)} + \frac{Cov_n((\bar{x} - \bar{x}_d), \Delta)}{Var_n(x)} \end{aligned} \quad (17)$$

where $\rho_{\bar{x}_d} \equiv \frac{Var_n(\bar{x}_j)}{\bar{x}_d(1 - \bar{x}_d)}$ is interclass correlation (ICC) of minority share across precincts in the district. If $\rho_{\bar{x}_d}$ is large (i.e., precincts are fully racially sorted, with percent minority close to 0 or 100 percent), the impact of correlations between majority voter behavior and minority status on aggregation bias will be minimal. Conversely if $\rho_{\bar{x}_d}$ is small (i.e., precinct minority shares are typically very close to the district average), those correlations will have much larger effects on measurement.

Derivation To derive Equation 17 we start with the standard decomposition of aggregation bias. Following Glynn et al. (2008),¹⁷ the Ecological Regression coefficient (in Equation 1) can be decomposed into the following components

$$\begin{aligned}
E[\bar{\Delta}_{eco,d}|\bar{x}_d] &= \bar{\Delta}_{n,d} + \overbrace{\frac{\sum_{j \in S_d} n_j (\bar{x}_j - \bar{x}) (\beta_i^w - \bar{\beta}^w)}{\sum_{j \in S_d} n_j (\bar{x}_j - \bar{x})^2}}^{\text{Intercept Bias}} \\
&\quad + \underbrace{\frac{\bar{x} \sum_{j \in S_d} n_j (\bar{x}_j - \bar{x}) (\Delta_i - \bar{\Delta})}{\sum_{j \in S_d} n_j (\bar{x}_j - \bar{x})^2} + \frac{\sum_{j \in S_d} n_j (\bar{x}_j - \bar{x})^2 (\Delta_i - \bar{\Delta})}{\sum_{j \in S_d} n_j (\bar{x}_j - \bar{x})^2}}_{\text{Slope Bias}} \\
&= \bar{\Delta}_{n,d} + \frac{Cov_n(x, \beta^w)}{Var_n(x)} + \bar{x}_d \frac{Cov_n(x, \Delta)}{Var_n(x)} + \frac{Cov_n((\bar{x} - \bar{x}_d), \Delta)}{Var_n(x)}
\end{aligned}$$

Next, we provide expression for $\bar{\Delta}_d$ (in Equation 18) that when subtracted from both sides of this equation yields Equation 17. First, consider the sample covariance between precinct composition (x_j) and minority voting behavior (β_j^b):

$$\begin{aligned}
Cov_n(x, \beta^b) &= \frac{1}{n_d} \sum_{j \in S_d} n_j (\bar{x}_j - \bar{x}_d) (\beta_j^b - \bar{\beta}_{n,d}^b) \\
&= \frac{1}{n_d} \sum_{j \in S_d} n_j (\bar{x}_j \beta_j^b - \beta_j^b \bar{x}_d - \bar{x}_j \bar{\beta}_{n,d}^b + \bar{\beta}_{n,d}^b \bar{x}_d) \\
&= \frac{1}{n_d} \sum_{j \in S_d} n_j \bar{x}_j \beta_j^b - \frac{\bar{x}_d}{n_d} \sum_{j \in S_d} n_j \beta_j^b - \frac{\bar{\beta}_{n,d}^b}{n_d} \sum_{j \in S_d} n_j \bar{x}_j + \bar{\beta}_{n,d}^b \bar{x}_d \\
&= \frac{1}{n_d} \sum_{j \in S_d} n_j \bar{x}_j \beta_j^b - \bar{\beta}_{n,d}^b \bar{x}_d
\end{aligned}$$

¹⁷Note that our discussion excludes a third source of ecological bias described by Glynn et al. (2008) – the correlation between precinct composition and individual confounders. They allow for individual-level covariates, z_o , that may be correlated with race (x_i) and vote choice ($Vote D_i$).

$$Vote D_i = \beta_{id}^w + \Delta_{id} \cdot x_i + \gamma z_i + \epsilon_i$$

For expositional simplicity, we focus on the special case where $\gamma = 0$ (see footnote 1).

Equivalently,

$$\begin{aligned}\sum_{j \in S_d} n_j \bar{x}_j \beta_j^b &= n_d \text{Cov}_n(x, \beta^b) + n_d \bar{x}_d \beta_{n,d}^b \\ \frac{\sum_{j \in S_d} n_j \bar{x}_j \beta_j^b}{n_d \bar{x}_d} &= \frac{1}{x_d} \text{Cov}_n(x, \beta^b) + \beta_{n,d}^b\end{aligned}$$

where $\bar{\beta}_{n,d}^b \equiv \frac{1}{n_d} \sum_{j \in S_d} n_j \beta_{jd}^b$ denotes the population-weighted minority party D support.

Likewise, it follows that

$$\begin{aligned}\sum_{j \in S_d} n_j (1 - \bar{x}_j) \beta_j^w &= n_d \text{Cov}_n(1 - \bar{x}, \beta^w) + n_d (1 - \bar{x}_d) \beta_{n,d}^w \\ &= n_d (1 - \bar{x}_d) \beta_{n,d}^w - n_d \text{Cov}_n(\bar{x}, \beta^w) \\ \frac{\sum_{j \in S_d} n_j (1 - \bar{x}_j) \beta_j^w}{n_d (1 - \bar{x}_d)} &= \beta_{n,d}^w - \frac{1}{1 - x_d} \text{Cov}_n(\bar{x}, \beta^w)\end{aligned}$$

where $\bar{\beta}_{n,d}^w \equiv \frac{1}{n_d} \sum_{j \in S_d} n_j \beta_{jd}^w$ is the population-weighted majority party D support

By definition, $\beta^b = \beta^w + \Delta$. Consequently,

$$\begin{aligned}\bar{\Delta}_d &= \frac{1}{x_d} \text{Cov}_n(x, \beta^b) + \beta_{n,d}^b + \frac{1}{x_d} \text{Cov}_n(\bar{x}, \beta^w) - \beta_{n,d}^w \\ &= \frac{1}{\bar{x}_d} \text{Cov}_n(x, \beta^b) + \frac{1}{1 - \bar{x}_d} \text{Cov}_n(\bar{x}, \beta^w) + \bar{\Delta}_{n,d} \\ &= \frac{1}{\bar{x}_d (1 - \bar{x}_d)} \text{Cov}_n(x, \beta^w) + \frac{1}{\bar{x}_d} \text{Cov}_n(\bar{x}, \Delta) + \bar{\Delta}_{n,d}\end{aligned}\tag{18}$$

C Simulation Details

Summary of Key Parameters

The parameters listed in this table provide the values used for our simulation. Each is explained in more depth in the following subsections.

Table 3: Simulation Parameters and Values

Parameter	Description	Default
<i>Population</i>		
J	Number of precincts	830
K	Number of urban centers	2
$\bar{N}$	Mean precinct population	2,300
σ_N	SD of precinct population	1,500
μ_m	State-wide minority share target	0.25
α	Beta distribution shape parameter	0.80
ρ_{NX}	Correlation between log-population and minority share	0.20
<i>Voting</i>		
μ_1, μ_0	Group mean Democratic vote probability	0.81, 0.37
σ_1, σ_0	Group-specific SD	0.10, 0.30
ρ_π	Within-precinct group correlation	0.20
<i>Segregation</i>		
ω	Weight on Democratic voteshare in composite score	0.40
λ	Weight on socioeconomic rank (0 = random, 1 = deterministic)	0.5 / 1.0
<i>Redistricting</i>		
D	Number of districts	14
ϵ	Population deviation tolerance	$\pm 1\%$
N_{plans}	Number of neutral ReCom steps	1,000
N_{ens}	Partisan ensemble size	1,000
B	Burst length (ReCom steps per burst)	10
T	Number of bursts per ensemble member	100
P	Patience (early stopping threshold)	6
k	Sigmoid steepness in expected seat scoring	60

Generating a Population

We sample precinct population values from a log-normal distribution

$$N_j \sim \text{LogNormal}(\mu_N, \sigma_{\ln}^2)$$

where $\sigma_{\ln}^2 = \ln(1 + \sigma_N^2 / \bar{N}^2)$ and $\mu_N = \ln(\bar{N}) - \sigma_{\ln}^2 / 2$ so that precinct population has mean $\bar{N}$ and standard deviation σ_N .

Minority share values are drawn from a Beta distribution

$$\bar{X}_j \sim \text{Beta}\left(\alpha, \alpha \cdot \frac{1 - \mu_m}{\mu_m}\right)$$

where μ_m is a state-wide minority share target and $\alpha < 1$ produces a right-skewed distribution with substantial mass near zero. Finally, we induce a minor correlation between precinct size and minority population share via

$$\bar{X}_j = \text{logit}^{-1}\left(\text{logit}(\bar{X}_j) + \rho_{NX} \cdot z_j^{(N)}\right)$$

where $z_j^{(N)}$ is the standardized log-population in unit j and ρ_{NX} is a correlation coefficient.

Examples of both distributions are shown in Figure 5.

C.0.1 Imposing Ground Truth Votes

Our vote model directly implements the assumptions of King's method for Ecological Inference by drawing correlated minority and majority voting probabilities within precincts jointly from a truncated normal distribution:

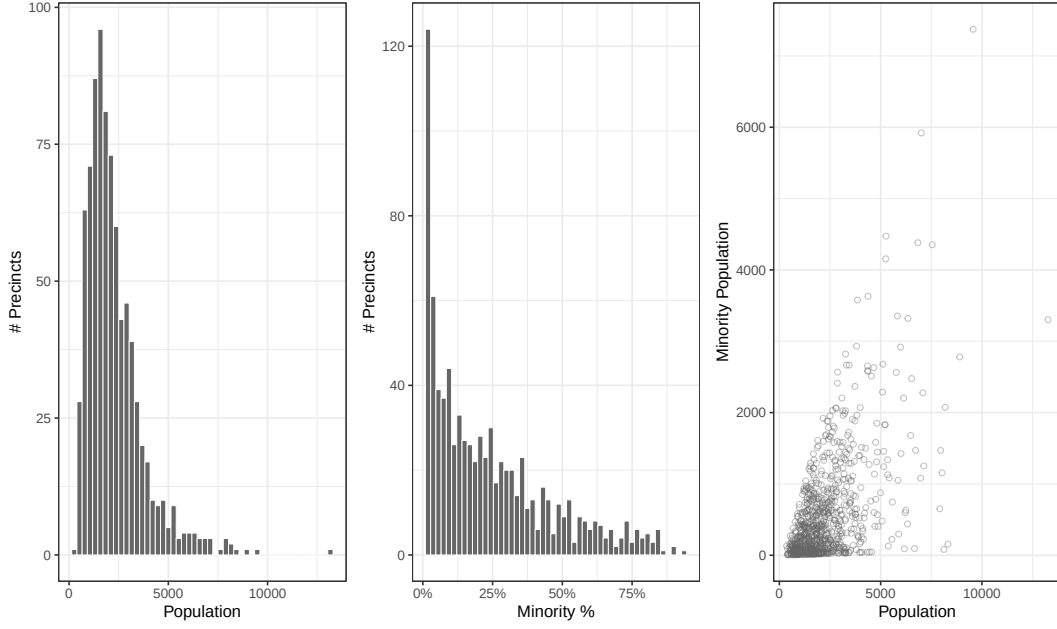


Figure 5: Distribution of Population and Minority Share at Precinct Level

$$\begin{pmatrix} \pi_{j,1} \\ \pi_{j,0} \end{pmatrix} \sim \text{TN}_{[0,1]^2} \left(\begin{pmatrix} \mu_1 \\ \mu_0 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho_\pi \sigma_1 \sigma_0 \\ \rho_\pi \sigma_1 \sigma_0 & \sigma_0^2 \end{pmatrix} \right) \quad (19)$$

where μ_1 and μ_0 are population means for minority and majority groups, $\sigma_1^2 < \sigma_0^2$ reflects cohesive voting behavior in the minority group, and ρ_π is the within-precinct correlation between minority and majority vote probabilities. Individual votes are then generated by binomial draws from these probabilities

$$V_{j,x} \sim \text{Binomial}(n_{j,x}, \pi_{j,x}) \quad (20)$$

for $x \in \{0, 1\}$, where $n_{j,x}$ is the number of voters in group x within precinct j . Importantly, vote probabilities are drawn once and votes are generated before the population is assigned to any precinct geography. All variation in the spatial distribution of voting behavior therefore arises from the segregation assignment mechanisms described below, not from any spatially correlated vote-generation process. This design choice ensures that any ecological bias we observe in redistricted maps is attributable to the interaction of residential sorting and mapmaker strategy, rather than to spatial dependence in the data-generating process.

Where precinct or VTD shapefiles are used, as in our analysis of North Carolina, we apply the same generative vote model to the census CVAP group-specific population counts. That is, we retain the real geography and demographic composition of each precinct but replace observed election returns with simulated votes drawn from Equation 19, using the same hyperparameters $(\mu_1, \mu_2, \sigma_1, \sigma_2, \rho_\pi)$ as in the synthetic case.

Spatial Grid Generation

We begin by constructing our state’s geography by sampling J points from a two-dimensional uniform plane. We employ rejection sampling where the point-wise probability of acceptance P is defined by a density function

$$P = \frac{\rho(x, y)}{\rho_{\max}} \quad (21)$$

$$\rho(x, y) = \rho_0 + \sum_{c=1}^K I_c \cdot m \cdot f(d_c) \quad (22)$$

The parameter ρ_0 defines a baseline density for rural areas. I_c is the intensity of urban center $c \in \{1, \dots, K\}$. Intensities are sampled $I_c \sim \text{U}(0.3, 1.0)$ and m is a scalar that

increases the density of points located in the center of urban areas. Finally,

$$f(d_c) = \begin{cases} 1 & \text{if } d_c < r \quad (\text{within city core radius}) \\ e^{-(d_c-r)^p / \delta^p} & \text{if } d_c \geq r \end{cases} \quad (23)$$

is a piece-wise function that takes the value of 1 when a grid point is within the specified radial distance from an urban center, and a value given by the kernel of the generalized normal distribution otherwise. The parameter p determines the shape of decay (gaussian when $p = 2$) as a function of the euclidean distance $d_c = \sqrt{(x - x_c)^2 + (y - y_c)^2}$ between a point and on the grid and an urban center c . δ determines the scale of decay. The summation term in equation 22 ensures that all urban centers influence the density of every point on the grid. The left-most panel of Figure 6 shows a generated set of points.

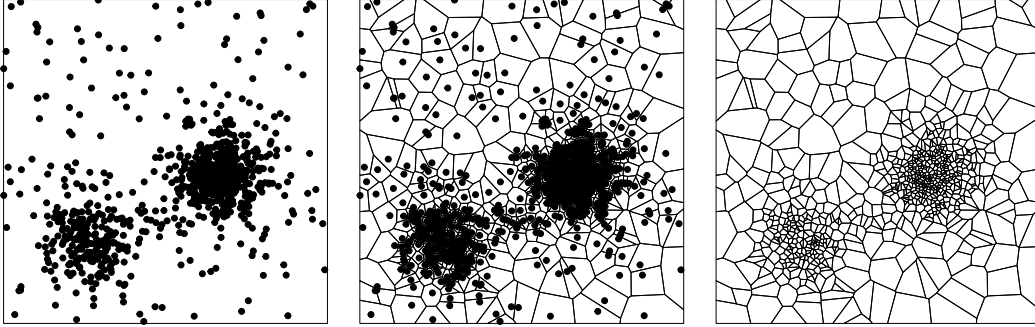


Figure 6: Creating Precinct Geographies

Next we use Voronoi tessellation to partition the euclidean space into N_j cells. Each sampled point serves as a seed for a cell, and cells contain all points for which the

minimum distance to any other cell is greater than the distance to itself. The middle panel of Figure 6 shows the sampled seed points and the precinct cells. The right-most panel shows just the precinct geography without the seed points. The result is a state-wide precinct-map that mimics the variation in size across rural and urban areas observed in real maps. Figure 6 uses two city centers and 830 precincts, the same values used for our main analysis. Although precinct-level populations are roughly equal in our simulation, urban precincts occupy a much smaller area, allowing populations to be more densely distributed in and around urban areas.

Inducing Residential Segregation

We assign our fixed population of voters to a fixed precinct geography, varying the assignment mechanism across three scenarios to vary the degree to which demographic (minority status) and partisan (vote choice) features cluster spatially. Let j index precinct populations and k index precinct geographic boundaries. Each level of segregation bias is a one-to-one mapping of precincts j to precinct boundaries k . For the medium and high scenarios, we match every j to a k based on the demographics and voting behavior of j ($\bar{X}_j, \bar{Y}_j$), and the urbanicity of precinct boundary k (U_k).

Low In the low segregation scenario, precinct populations j are randomly assigned to precinct geographic boundaries k , creating a state with near-zero spatial correlation in minority population share (top-left panel of Figure 7) or Democratic voteshare (bottom-left panel of Figure 7).

Medium and High In the medium segregation scenario, we construct a composite socioeconomic score

$$S_j = \omega \cdot \bar{Y}_j + (1 - \omega) \cdot \bar{X}_j$$

where $\bar{Y}_j, \bar{X}_j$ are the min-max standardized Democratic voteshare and minority population share respectively and ω is a weight. Assignment ranks are computed as a convex combination of socioeconomic and random ranks

$$R_j = \lambda \cdot \text{rank}(-S_j) + (1 - \lambda) \cdot \text{rank}(u_j), \quad u_j \sim \text{Uniform}(0,1)$$

and we set $\lambda = 0.5$ for medium segregation and $\lambda = 1$ for high segregation. Precincts j are assigned to precinct boundaries k by matching the rank order of R_j to the urbanicity-sorted boundaries. See Table 4 for an overview of the kinds of spatial autocorrelation in race and partisanship these levels create.

Segregation	Simulated		Real States	
	Minority Share	Dem. Vote Share	Minority Share	Dem. Vote Share
Low	0.023	0.025	0.401	0.893
Medium	0.348	0.294	0.610	0.787
High	0.814	0.625	0.702	0.723

Table 4: Moran’s I Measures of Spatial Autocorrelation

Note: Moran’s I statistics for spatial autocorrelation of minority population share and Democratic presidential vote share, computed using queen contiguity weights with row standardization. Simulated data use North Squarolina precincts generated at three levels of residential racial sorting. Real state comparisons use 2020 Census block populations (Black alone, P.L. 94-171) and 2024 presidential election precinct returns: Colorado (low segregation), North Carolina (medium), and Alabama (high).

C.0.2 Drawing Maps and Mapmaker Objectives

For each segregation scenario, we generate three sets of redistricting plans: a neutral ensemble and two partisan ensembles (one favoring Democrats and one favoring Republicans). All plans are represented as partitions over a graph of nodes (precincts)

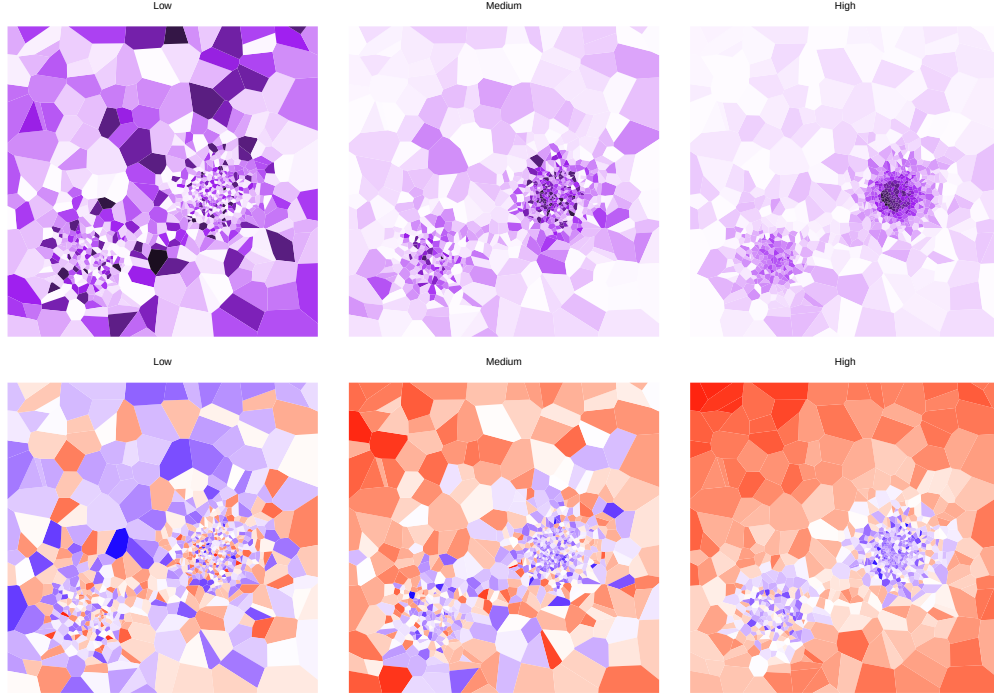


Figure 7: Levels of Residential Segregation

connected by edges (spatial contiguity), constructed using the MCMC framework implemented in the GerryChain Python library (DeFord, Duchin and Solomon, 2021). Plans in all three ensembles must satisfy three constraints: contiguity, meaning all districts must be connected, population balance, meaning each district has a $\pm 1\%$ population deviation tolerance, and compactness, meaning the number of cut edges¹⁸ is no more than triple the initial value.

Neutral Redistricting

Neutral plans are generated by a standard Recombination Markov chain (DeFord, Duchin and Solomon, 2021). Starting from a random initial partition of precincts into

¹⁸Cut edges are edges between two nodes in the graph that are assigned to different districts.

$D = 14$ districts, the chain performs a random walk over the space of valid redistricting plans. At each step, two adjacent districts are randomly selected, their nodes are merged into a single subgraph, and a new spanning tree is drawn. Cutting a randomly selected edge of this tree produces two new districts. If the resulting plan satisfied all three constraints listed above, it is accepted. We run N_{plans} steps and retain every accepted plan.

Partisan Redistricting

Partisan gerrymander ensembles are generated using short-burst optimization. The optimization mimics the will of an “omniscient” mapmaker: one who observes the true group-specific vote totals in each precinct, but is uncertain about the outcome of the next election. In drawing maps, the mapmaker asks “if a new election were held under this map, how many seats would my party expect to win?”

Each of the N_{ens} plans in the ensemble is produced by an independent optimization, proceeding as follows. Starting from an initial partition, we run T consecutive bursts of B Recombination steps each. Within each burst, the chain proposes and accepts plans exactly as in the neutral case. At the end of each burst, we evaluate the best plan found thus far using a measure of the expected seat count

$$\mathcal{S}(\mathbf{p}) = \sum_{d=1}^D \frac{1}{1 + \exp\left(-k \cdot \frac{v_d(\mathbf{p}) - 0.5}{\hat{\sigma}_d(\mathbf{p})}\right)}$$

where $v_d(\mathbf{p})$ is the target party’s vote share in district d under plan $\mathbf{p}$, k controls the steepness of the sigmoid, and $\hat{\sigma}_d(\mathbf{p})$ is an estimate of the standard deviation of the district-level vote share. $\mathcal{S}$ approximates the expected number of seats won under electoral uncer-

tainty. For example, a district where the mapmaker’s party won 55% of the vote with heterogeneous group behavior at the precinct level might be as risky as a district where the party won 52% in a low variance environment. The best scoring plan from each burst seeds the next burst and the final output of each optimization burst is the single best-scoring plan across all bursts. This process generates an ensemble of N_{ens} independently optimized plans for each party and segregation scenario.

C.1 Overview of Electoral Outcomes

Appendix Table 5 reports key district- and map-level characteristics across the segregation and gerrymandering parameters discussed above. Our simulations produce broadly Democratic maps, with Democrats winning more than half of the available seats on average across most scenarios and never winning fewer than 5 (of 14) seats. For the purposes of thinking about generalizability, our results might do well to characterize maps where gerrymandering can take a chamber from an even and potentially contentious split to a solid Democratic majority, but not maps where gerrymandering might swing the state from overwhelming Democratic to overwhelming Republican control.

As the first panel of Table 5 indicates, the degree of segregation constrains the range of feasible district characteristics. In an environment in which voters of different races are evenly distributed across space, for example, the expected win margin is small (no more than five percent in 38-40% of districts) and varies little across plans (standard deviation is five-seven percent). The majority of simulated districts favor Democrats by more than five percentage points, while fewer than three percent favor Republican by similar margins if segregation is minimal. By contrast, expected win margins are larger and more variable in electoral environments featuring greater residential segregation. Fewer than ten percent of districts produce narrow win margins in the high segregation scenario; as

many as 47 percent (52%) of districts favor Republican (Democratic) candidates by more than five percentage points. Likewise, majority-minority districts are absent under low segregation, but account for 1.5-3.5 percent of districts in scenarios with at least some segregation.

Next, we turn to how redistricting and residential segregation jointly affect the risk of vote dilution by race.¹⁹ The third panel of Table 5 reports the racial disparity in wasted votes, calculated as the difference of two ratios: $(\text{Prob}(\text{Wasted} \ \& \ \text{Minority}) / \text{Prob}(\text{Minority})) - (\text{Prob}(\text{Wasted} \ \& \ \text{Majority}) / \text{Prob}(\text{Majority}))$. Greater values indicate that minority voters are either disproportionately packed into safe districts won by co-partisan representatives or excessively cracked across districts electing non co-partisans. We find that racial disparities are most striking in Republican-leaning maps and in simulations where voters are racially and politically integrated. One common theme across segregation scenarios is that mapmakers' choices affect the risk of dilution for minority voters much more than for majority voters. For instance, the proportion of minority voters who are wasted may double, or even triple, when moving from pro-Democratic to pro-Republican maps, while the corresponding proportion of majority voters shifts by no more than 40%.

¹⁹To be sure, the most appropriate method for defining what it means for minority or majority voters to be "better off" or "worse off" in any particular electoral map is not straightforward. Some have argued that minority representation may be best achieved if those voters are concentrated in a few districts where they might translate their votes into seats. We might also think, and indeed courts have recognized, that such districts may insulate incumbents from electoral competition, undermining electoral incentives and responsiveness (Griffin, 2006).

Table 5: Electoral Features of Simulated State Across Segregation and Redistricting Levels

Seg.	Redist.	N	Comp. (%)	SD (%)	Safe D (%)	Safe R (%)	D Seats (Avg)	Min-Max D Seats	Maj-Min Dists	Waste Disp. (pp)
Low	Dem.	14000	70.9	5.2	18.0	11.1	12.4	11-13	0.0	-28.4
	Net.	14000	57.9	5.7	29.9	12.3	9.4	7-12	0.0	-11.6
	Rep.	14000	61.2	7.6	30.6	8.2	4.5	3-7	0.0	15.3
High	Dem.	14000	24.5	25.1	53.7	21.8	10.9	9-13	0.8	-20.1
	Net.	14000	57.9	5.7	29.9	12.3	9.4	7-12	0.0	-11.6
	Rep.	14000	21.4	25.9	36.4	42.2	5.1	4-7	1.2	6.0

Note: N = number of districts; Comp. = competitive districts (margin < 5%); SD = standard deviation of win margins; Safe D/R = safe Democratic/Republican districts (margin > 5%); D Seats (Avg) = average Democratic seats per map; Min-Max D Seats = minimum and maximum Democratic seats across all maps in scenario; Maj-Min Dists = average number of majority-minority districts per map; Waste Disp. = waste disparity in percentage points (minority waste rate minus majority waste rate), calculated as the population-weighted difference in waste rates. Higher segregation reduces competitiveness by creating more homogeneous districts.

D Additional Results

Table 6: Impact of Redistricting on ER/EI Performance (High Segregation Simulation)

	Ecological Regression		Ecological Inference	
	Min.	Maj.	Min.	Maj.
Neutral Map	-28.958*** (0.190)	8.863*** (0.053)	-17.816*** (0.175)	6.545*** (0.051)
Democratic Map	-2.726*** (0.282)	0.113 (0.085)	-0.579** (0.262)	-0.054 (0.079)
Republican Map	-4.195*** (0.279)	2.143*** (0.071)	-1.401*** (0.252)	1.685*** (0.067)
Competitive District	-3.602*** (0.546)	-1.207*** (0.119)	-4.595*** (0.491)	-0.984*** (0.119)
× Democratic Map	11.369*** (0.692)	-2.113*** (0.155)	7.390*** (0.600)	-1.778*** (0.145)
× Republican Map	2.162*** (0.708)	-2.491*** (0.151)	-0.135 (0.665)	-2.231*** (0.151)
Observations	42000	42000	42000	42000
R ²	0.012	0.031	0.007	0.027

Note: The dependent variable in all models is the prediction error for minority (Columns 1,3) or majority (Columns 2,4) vote share from ER or EI. Standard errors clustered at the map level are shown in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Electoral Features of North Carolina Across Redistricting Levels

Seg.	Redist.	N	Comp. (%)	SD (%)	Safe D (%)	Safe R (%)	D Seats (Avg)	Min-Max D Seats	Maj-Min Dists	Waste Disp. (pp)
Democratic	14000	43.8	8.3	42.0	14.2	12.0	11-13	0.0	-25.2	
Neutral	14000	37.5	9.0	47.4	15.1	9.5	7-12	0.0	-14.8	
Republican	14000	39.0	10.4	45.0	15.9	6.4	5-9	0.0	-1.6	

Note: N = number of districts; Comp. = competitive districts (margin < 5%); SD = standard deviation of win margins; Safe D/R = safe Democratic/Republican districts (margin > 5%); D Seats (Avg) = average Democratic seats per map; Min-Max D Seats = minimum and maximum Democratic seats across all maps in scenario; Maj-Min Dists = average number of majority-minority districts per map; Waste Disp. = waste disparity in percentage points (minority waste rate minus majority waste rate), calculated as the population-weighted difference in waste rates. Higher segregation reduces competitiveness by creating more homogeneous districts.